Quantifying Financial Repression in Emerging Markets*

Mert Onen†

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Abstract

Financial repression has historically accompanied periods of high fiscal needs. Yet there is no high-frequency measure of repression for emerging markets (EMs), where weaker institutions and lower debt tolerance make it especially prevalent. To that end, I examine the Domestic Bond Premium (DBP): the yield spread between EM local currency (LC) government bonds and AAA-rated supranational bonds in the same currency and tenor. Using data from 4,017 supranational bonds across 11 EM currencies, I document that the DBP is frequently negative and remains stable during periods of financial stress when government LC yield spreads over US Treasuries widen sharply. A simple portfolio choice model decomposes the DBP into default, liquidity, and financial repression components. The empirical evidence supports the model predictions by showing that default risk and liquidity alone cannot account for the low DBPs, pointing to repression as a key force bringing government borrowing costs below the supranational benchmarks.

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†University College London. Email: mert.onen.24@ucl.ac.uk.

1 Introduction

Global public debt rose sharply in the aftermath of the Global Financial Crisis (GFC) and again during the pandemic, reaching an estimated 95.1% of GDP in 2025 (Reis, 2026). This increase has renewed concern that governments may once again use financial repression, defined as policies that generate inelastic domestic demand for public debt, to contain their debt-service costs. Emerging markets (EMs), where public debt stands at its highest level in half a century (Kose et al., 2022), should be central to this renewed debate for several reasons.

In 2025 alone, emerging economies faced an \$8.2 trillion refinancing need (Institute of International Finance, 2025). Domestic banks are already heavily exposed to sovereign debt, with government securities accounting for 17% of their total assets in the aftermath of the pandemic (International Monetary Fund, 2022). Although EM public debt-to-GDP ratios are often below those of advanced economies (AEs), this comparison overstates their fiscal space. Higher and more volatile borrowing costs, more fragile market access, and shallower domestic capital markets mean that refinancing pressures can become acute at lower debt ratios. Moreover, where safeguards separating prudential regulation from sovereign financing are weak, policymakers have greater scope to use reserve requirements, holdings mandates, preferential regulatory and tax treatment, moral suasion, and capital account restrictions to redirect domestic savings toward government debt.

Yet there is no high-frequency, market-based measure of the intensity of financial repression across EMs. The main difficulty is that the borrowing cost that would prevail without policy-induced 'captive' demand is unobserved. The empirical object of interest is therefore the wedge between actual government borrowing costs ($y^{\text{Gov}}_{j,t,\tau}$) and this unobserved counterfactual ($y^{\text{Gov, no repression}}_{j,t,\tau}$):

$$\text{Repression wedge}_{j,t,\tau} \equiv y^{\text{Gov}}_{j,t,\tau} - y^{\text{Gov, no repression}}_{j,t,\tau},$$

where j indexes the currency, t the date, and τ the tenor. A suitable benchmark is therefore needed for the counterfactual yield.

This paper uses AAA-rated supranational bonds issued in EM local currencies as that benchmark.[1] These bonds provide a good proxy because they are denominated in the same currency and can be matched by tenor to domestic government bonds, but are issued offshore by highly creditworthy institutions that are outside the purview of domestic financial regulation. Given this, I define the Domestic Bond Premium (DBP) as

$$DBP_{j,t,\tau} \equiv y^{\text{Gov}}_{j,t,\tau} - y^{\text{Sup}}_{j,t,\tau},$$

where $y^{\text{Sup}}_{j,t,\tau}$ represents the yield on supranational bonds in the same currency and tenor. Because the two yields share identical currency exposure, the DBP nets out FX risk by construction. After accounting for sovereign default risk and differential liquidity, a lower DBP indicates that government borrowing costs are compressed relative to comparable default-free benchmarks.[2]

[1] Supranational financial institutions are multilateral development banks owned and funded by several governments to finance development projects globally. For details on supranational issuance in EM currencies, see Dao and Gourinchas (2026).

[2] Throughout this paper, the term 'default-free' refers specifically to the absence of outright default (i.e., nominal repayment) risk, reflecting supranational issuers' high creditworthiness. These bonds remain subject to

Figure 1: Five-year Turkish lira government and supranational zero-coupon yields

Notes: This figure plots five-year Turkish lira denominated zero-coupon rates for government bonds (blue) and supranational bonds (orange) in percent. The shaded area marks Turkey's most acute financially repressive episode, from the December 2021 introduction of FX-protected lira deposits (KKM) to the May 2024 removal of the Securities Maintenance Practice (SMP); the green dashed lines bracket the subperiod when SMP requirements were most stringent (May 2022–June 2023). Sources: Bloomberg, Refinitiv, and author's calculations.

Turkey provides a striking illustration of how negative this wedge can get. As Figure 1 shows, during Turkey's most acute repressive episode (the shaded area) the five-year lira DBP reached roughly −20 percentage points, even as annualised inflation surpassed 80% and CDS premia spiked above 800 basis points, conditions that, on standard grounds, would raise rather than compress government yields. I examine this episode in detail in Section 5.5. The Turkish case is the most extreme example in my sample, but it is not isolated: highly negative DBPs are a common feature across many EMs.

To document this more systematically, I construct a novel dataset of supranational zero-coupon yield curves for 11 EM currencies using data from 4,017 supranational bonds issued between 1997 and 2025 by eight supranational financial institutions (for details, see Section 3). Across this panel, the DBP is persistently low and frequently negative, meaning EM governments in my sample are often able to borrow at or even below default-free benchmarks in their own currencies.

To formalise the financial repression interpretation, Section 2 presents a simple asset pricing model in which domestic investors are subject to a minimum-holdings constraint. When the constraint binds, forced demand props up government bond prices and opens a financial repression wedge in yields. The model decomposes EM LC bond yields into a shadow risk-free rate, a default premium, a liquidity premium, private convenience services, and this repression wedge. Adding foreign investors to the model delivers the key result: the repression wedge increases with the domestic ownership share of the government bond market, because captive domestic

currency and liquidity risk.

institutions cannot escape the constraint, whereas unrepressed and flighty foreign investors can discipline the price.

Importantly, repression has two effects that pull in opposite directions. Forced demand raises government bond prices, but when the policy is expected to persist, it may also distort the marginal value of investors' wealth and with it the shadow risk-free rate in the economy. The net effect of repression on the level of government yields is therefore theoretically ambiguous. The DBP resolves this identification problem. Because both the government and supranational bonds are denominated in the same currency, the risk-free rate and convenience components cancel in the spread; conditional on observable proxies for default risk and liquidity, movements in the DBP within a currency and tenor hence trace movements in repression intensity. The framework also disciplines the empirics with sign predictions: rising default risk should widen the DBP, deteriorating supranational liquidity should compress it, and intensifying financial repression should compress it further, potentially below zero.

The aggregate patterns in DBP are consistent with these theoretical predictions. In addition to being persistently low and frequently negative, the DBP exhibits stability during stress episodes such as the GFC or the Covid-19 recession, when EM LC yield spreads over US Treasuries surge. This insensitivity to deteriorating financial conditions is particularly pronounced at shorter maturities, where the DBP declines or turns negative during crises. Such patterns are consistent with financial repression: captive domestic demand sustains government bond prices even as fundamentals deteriorate.

Beyond these aggregate patterns, I provide four pieces of evidence tying the DBP to financial repression. First, the variation in the DBP is predominantly country-specific. A variance decomposition of the panel dataset shows that between 80-90% of the DBP variation stems from persistent country characteristics and time-varying local factors, contrasting with the strong commonality found in foreign-currency yields (González-Rozada and Yeyati, 2008; Du and Schreger, 2016) or credit default swap (CDS) spreads (Longstaff et al., 2011). The idiosyncratic nature of the variation in DBP points to local financial conditions and domestic policies rather than global risk sentiment as the primary drivers.

Second, in panel regressions, proxies for financial repression are statistically and economically significant in explaining DBP movements across all specifications. Sovereign default risk, by contrast, is significant only at longer maturities, consistent with the low probability of outright nominal default over short horizons. Interacting default risk with domestic ownership reveals that the positive association between CDS spreads and the DBP is substantially attenuated when the domestic investor base is large, consistent with the model's prediction that captive demand insulates government borrowing costs from deteriorating creditworthiness. These results are robust to currency–tenor–year fixed effects that absorb slow-moving unobservables, and to replacing the domestic holdings with the reserve requirement ratio, a direct policy lever that can tilt bank portfolios toward domestic government debt.

Third, the DBP responds at high frequency to discrete policy actions. Using daily local projections around macroprudential policy events extracted from the IMF's Integrated Macroprudential Policy database (iMaPP), I find that the DBP compresses following events predicted to expand captive demand for LC government bonds and widens following events predicted to reduce it. Meanwhile, the DBP is largely unresponsive to events that are not expected to affect domestic demand for government bonds.

Finally, the two narrative case studies in Section 5.5 illustrate how financial repression can work through different mechanisms. In Turkey, shown in Figure 1, repression was deliberately imposed to control borrowing costs through banking regulation. In Russia, by contrast, an exogenous geopolitical shock made repressive mechanisms exceptionally effective: following the war in 2022, international sanctions drove out foreign investors while capital controls trapped domestic institutions in LC government bonds, holding government yields well below supranational rates despite a substantial rise in Russian sovereign default risk.

Together, these results support the use of DBP as a high-frequency, market-based measure for tracking financial repression intensity across countries. The prevalence of negative DBPs demonstrates that financial repression can compress EM government borrowing costs below those of AAA-rated supranational institutions. While these negative spreads superficially resemble the convenience yields documented for US Treasuries, they are more likely to arise from EM governments' ability to generate inelastic domestic demand for their liabilities rather than global demand for safe assets. This distinction matters for debt sustainability analysis: unlike safe-haven assets, captive demand for EM LC government bonds can unwind rapidly when regulations loosen or the banking sector comes under stress.[3]

Relation to literature. This paper relates to several strands of literature. First, I build on the work of Du and Schreger (2016) who quantify the non-currency risk premium in EM LC sovereign yield spreads; these authors examine the yield spread between EM LC sovereign bonds and 'synthetic risk-free rates', defined as the US Treasury yield plus the relevant cross-currency swap rate. However, their approach also incorporates the US Treasury premium and CIP deviations.[4] Importantly, while Du and Schreger (2016) were the first to measure the government-supranational spread for two countries using limited supranational bond data, a systematic EM-wide analysis of this spread and its relation to financial repression has been absent from the literature.

From a methodological perspective, this paper relates to studies that use supranational bonds to isolate specific yield spread components. Dao and Gourinchas (2026) use supranational bonds to estimate 'purified' CIP deviations that partial out EM credit risk. Schwarz (2019) uses the difference between German Bund and KfW euro yields to estimate a model-free measure of market liquidity for the euro area during the Global Financial Crisis (GFC) and the European debt crisis.

My findings also connect to the safe-asset demand and convenience yield literature. Convenience yields arise when investors accept lower financial returns on certain assets in exchange for non-pecuniary benefits such as exceptional liquidity, universal acceptance as collateral, and

[3]For a discussion of how rapidly unwinding financial repression can trigger crises, see Diaz-Alejandro (1985) on Latin America. For the sovereign-bank nexus, see Acharya et al. (2014), Gennaioli et al. (2014), and Farhi and Tirole (2018).

[4]Covered Interest Parity (CIP) is a no-arbitrage condition requiring that interest rate differentials between two currencies equal the forward premium. In other words, borrowing in one currency, converting to another, and hedging exchange rate risk via forward contracts should yield the same return as borrowing directly in the second currency. Deviations from CIP may reflect excess demand for dollar hedging or limited supply of dollar forwards (Du and Schreger, 2022a). Also see Appendix B.2.

safe-haven properties. Krishnamurthy and Vissing-Jorgensen (2012), Greenwood et al. (2015), Nagel (2016), Du et al. (2018), and Jiang et al. (2021) document how US Treasury securities command such premiums over comparable assets or over what standard asset-pricing models would predict. I show that EM LC government bonds can also trade at a premium to default-free assets, as evidenced by negative DBPs. However, rather than reflecting genuine investor preference for safety and liquidity, EM convenience premiums potentially arise from regulatory pressure on domestic financial institutions to hold government debt.

In this respect, this paper is strongly grounded in the financial repression literature. The term originates with McKinnon (1973) and Shaw (1973), who characterised how policies such as interest rate ceilings and reserve requirements channel domestic savings toward the government at below-market rates, with adverse consequences for capital accumulation. Giovannini and de Melo (1993) provided static estimates of the implicit tax rate that governments impose on its debtholders by suppressing local interest rates below foreign rates across 24 developing economies. Reinhart and Sbrancia (2015) document how advanced economy governments historically extracted resources from domestic financial systems through negative real interest rates alongside financial repression, estimating that after World War II such policies reduced government debt burdens significantly.

In the modern era, Becker and Ivashina (2018) and Ongena et al. (2019) provide evidence of similar mechanisms during the European debt crisis, where regulatory pressure and moral suasion led banks to increase domestic government bond holdings despite deteriorating sovereign creditworthiness. Broner et al. (2014) offer a complementary perspective, documenting that sovereign bond markets in turbulent times are characterised by creditor discrimination: domestic investors that are subject to local regulations absorb an increasing share of government debt as foreign investors withdraw amid rising credit risk. This mechanism is central to the theoretical framework developed in Section 2.2, where the financial repression premium on government debt increases as the domestic ownership share rises.

On the theory side, Reis (2021a) shows that financial repression can expand fiscal capacity through a wedge between the return on private capital and the interest rate on government debt, but that this comes at the cost of misallocating resources away from productive investment. In related work, Reis (2021b) models macroprudential policy as the share of government bonds that banks are required to hold and shows that tighter regulation raises bond prices along a downward-sloping demand curve, leaving a 'fiscal footprint' that lowers government borrowing costs; he stresses that once a fiscally constrained authority exploits this footprint, the line separating macroprudential policy from financial repression becomes a thin one. Chari et al. (2020) show that financial repression, though not optimal under commitment, can serve as a costly credibility mechanism: by forcing banks to hold government debt, governments make default more damaging to the domestic financial system and can thereby sustain higher debt levels. More recently, Itskhoki and Mukhin (2025) study financial repression in the currency market under sanctions, comparing it with conventional policies such as FX interventions. They show that although repression can be used to reallocate resources and extract fiscal surplus under financial isolation, it is generally welfare-reducing as a stabilisation tool.

As Kose et al. (2022) note, EMs face increasingly limited options to manage their rapidly rising public debt levels and interest burdens since the pandemic. This constrained fiscal environment makes it increasingly relevant to study financial repression. A natural starting point is to test whether repression is systematically reflected in the pricing of EM local-currency sovereign bonds across countries and over time. The DBP provides a high-frequency, market-based measure for that purpose.

The rest of the paper proceeds as follows. Section 2 develops the theoretical framework and decomposes the DBP into default risk, relative liquidity, and financial repression premia. Section 3 describes the data and construction of supranational yield curves. Section 4 presents stylised facts on the DBP. Section 5 presents the empirical evidence. Section 6 concludes.

2 Framework

2.1 Environment and domestic investor problem

This section introduces a portfolio choice model that decomposes bond yields and the DBP into distinct risk premium components. In this framework, a representative domestic investor chooses the quantity of two one-period LC zero-coupon bonds:[5] (i) a government bond with outright-default risk, and (ii) a supranational bond that is effectively default-free. Bond purchases are subject to transaction costs, interpreted as liquidity frictions.[6] Moreover, holding either bond provides *private* convenience services (e.g., use as collateral in interbank markets). Given both bonds are in the same currency and maturity, I assume these services do not depend on the identity of the issuer. Rather, they are a function of total bond holdings, $T_{t+1} \equiv B^{D,Gov}_{t+1} + B^{D,Sup}_{t+1}$, where superscripts index the investor (D for domestic, F for foreign) and the bond (Gov for government, Sup for supranational).[7] Crucially, I use a minimum-holdings constraint on government bonds as a reduced-form representation of policies that generate inelastic domestic demand for sovereign debt such as holdings mandates, preferential tax or regulatory treatment of sovereigns, moral suasion, and restrictions on alternative assets.

Preferences and budget constraint. The domestic investor chooses consumption c_t and bond holdings $B^{D,Gov}_{t+1}$ and $B^{D,Sup}_{t+1}$ to maximise the expected discounted utility

$$U_0 = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big[u(c_t) + v\big(T_{t+1}\big) \Big], \tag{1}$$

where $\beta \in (0, 1)$, $u'(c) > 0$, $u''(c) < 0$, and v captures the private convenience services from holding LC bonds, with $v' > 0$ and $v'' < 0$. At time t, a government bond with face value one trades at price P^{Gov}_t and pays $1 - \delta_{t+1}$ units of LC at $t + 1$, where $\delta_{t+1} \in [0, 1]$ is the random and exogenous sovereign haircut. The supranational bond trades at P^{Sup}_t and pays one unit of

[5] The one-period setup is for clarity of exposition. Extending the model to include multi-period zero-coupon bonds yields an analogous decomposition for each tenor, leaving the qualitative predictions unchanged. The subsequent empirical analysis uses zero-coupon yields at 1, 3, 5, and 10-year tenors.

[6] Allowing for other tradable assets such as equities or USD-denominated bonds would result in additional Euler equations but would not change the DBP decomposition, which is the main object of interest.

[7] The issuer-symmetry of private convenience services is an approximation. Section 2.3 discusses how the empirical implementation accommodates a departure from this assumption.

LC at $t+1$ almost surely. Asset purchases are subject to per-unit transaction costs $\varphi_t^{Gov} \geq 0$ and $\varphi_t^{Sup} \geq 0$. Each period, the investor receives a deterministic endowment $\omega > 0$. As such, the per-period budget constraint of the investor is

$$c_t + \left(P_t^{Gov} + \varphi_t^{Gov}\right) B_{t+1}^{D,Gov} + \left(P_t^{Sup} + \varphi_t^{Sup}\right) B_{t+1}^{D,Sup} = \omega + (1-\delta_t) B_t^{D,Gov} + B_t^{D,Sup}, \tag{2}$$

where the left-hand side shows expenditures and the right-hand side available resources, or total wealth, at time t. Due to sovereign default risk, an investor receives the face value of their government bond holdings carried over from time $t-1$ only if $\delta_t = 0$. The sovereign haircut δ_t, the financial repression parameter κ_t, and the transaction costs $\varphi_t^{Gov}, \varphi_t^{Sup}$ are observed at time t but their future values are unknown.

Financial repression. The model represents financial repression with a minimum-holdings constraint. This reduced-form device captures the common demand effect of policies that directly or indirectly tilt portfolios toward government debt:

$$P_t^{Gov} B_{t+1}^{D,Gov} \;\geq\; \kappa_t W_t, \qquad W_t \equiv \omega + (1-\delta_t) B_t^{D,Gov} + B_t^{D,Sup}, \tag{3}$$

where $\kappa_t \in [0,1]$ indexes the intensity of repression. A higher κ_t represents more intensive financial repression, and when the constraint binds, it generates additional domestic demand for government bonds.

Asset prices. Let $\mu_t = u'(c_t) > 0$ denote the Lagrange multiplier on the budget constraint and let $\lambda_t \geq 0$ denote the multiplier on the minimum-holdings constraint (shadow cost of financial repression). Define the *constrained* pricing kernel[8] and financial repression wedge

$$\tilde{M}_{t+1} \equiv \beta \, \frac{u'(c_{t+1}) - \lambda_{t+1}\kappa_{t+1}}{u'(c_t)}, \qquad \rho_t \equiv \frac{\lambda_t}{u'(c_t)} \geq 0, \tag{4}$$

and the marginal convenience of LC bonds in marginal utility terms

$$\chi_t \equiv \frac{v'\left(T_{t+1}\right)}{u'(c_t)}, \tag{5}$$

which is decreasing in total domestic LC holdings T_{t+1}. Then, the investor's optimality conditions imply the following pricing equations (see Appendix A for all derivations):

$$P_t^{Sup} + \varphi_t^{Sup} = \mathbb{E}_t\left[\tilde{M}_{t+1}\right] + \chi_t, \tag{6}$$

$$P_t^{Gov} + \varphi_t^{Gov} = \mathbb{E}_t\left[\tilde{M}_{t+1}(1-\delta_{t+1})\right] + \chi_t + \rho_t P_t^{Gov}. \tag{7}$$

Equation (7) highlights the role of financial repression: keeping other components fixed, when the constraint binds ($\rho_t > 0$), the policy-induced demand raises the price of government bonds.

[8] I assume $u'(c_{t+1}) - \lambda_{t+1}\kappa_{t+1} > 0$ almost surely, which ensures the positivity of the constrained pricing kernel.

Interest rates. Define the shadow risk-free rate as

$$\tilde{r}_t \equiv -\log \tilde{q}_t, \qquad \text{where} \qquad \tilde{q}_t \equiv \mathbb{E}_t\big[\tilde{M}_{t+1}\big], \tag{8}$$

and let $D_t \equiv \mathbb{E}_t[\tilde{M}_{t+1}\delta_{t+1}]$. Using (6)–(7) and a log-linear approximation around a frictionless, default-free benchmark, the continuously compounded one-period zero-coupon bond yields satisfy

$$y_t^{Sup} \approx \tilde{r}_t + \underbrace{\frac{\varphi_t^{Sup}}{\tilde{q}_t}}_{L_t^{Sup}} + \underbrace{\Big(-\frac{\chi_t}{\tilde{q}_t}\Big)}_{R_t^{Sup}}, \tag{9}$$

$$y_t^{Gov} \approx \tilde{r}_t + \underbrace{\frac{\mathbb{E}_t[\tilde{M}_{t+1}\delta_{t+1}]}{\tilde{q}_t}}_{I_t^{Gov}} + \underbrace{\frac{\varphi_t^{Gov}}{\tilde{q}_t}}_{L_t^{Gov}} + \underbrace{\Big(-\frac{\chi_t}{\tilde{q}_t} - \rho_t\Big)}_{R_t^{Gov}}. \tag{10}$$

Let $b \in \{Gov, Sup\}$ index the bond type and j index the currency. The continuously compounded yield on bond b issued in currency j at time t can thus be written as

$$y_{j,t}^b \approx \underbrace{\tilde{r}_{j,t}}_{\substack{\text{shadow}\\ \text{risk-free rate}}} + \underbrace{I_{j,t}^b}_{\substack{\text{default risk}\\ \text{premium}}} + \underbrace{L_{j,t}^b}_{\substack{\text{liquidity}\\ \text{premium}}} + \underbrace{R_{j,t}^b}_{\substack{\text{convenience and}\\ \text{repression}\\ \text{premium}}}. \tag{11}$$

The shadow risk-free rate $\tilde{r}_{j,t}$ represents the theoretical return on a default-free and frictionless asset. The remaining components are premiums above this baseline. The default risk premium $I_{j,t}^b$ compensates investors for expected losses from default; it increases when issuer creditworthiness deteriorates. The liquidity premium $L_{j,t}^b$ reflects compensation for trading costs; it rises when bid-ask spreads widen or trading volume declines. $R_{j,t}^{Sup}$ reflects the private convenience premium on LC bonds. $R_{j,t}^{Gov}$ reflects the same convenience premium and an active financial repression wedge. Therefore, when the financial repression constraint binds ($\rho_{j,t} > 0$), government bonds can trade at a premium relative to default-free but otherwise similar assets denominated in the same currency. Because the private convenience premium is common to the two bonds, it will difference out of the spread between them; Section 2.3 develops this point.

2.2 Effectiveness of financial repression

When does the financial repression constraint bind and thus reduce borrowing costs? To answer this question, I extend the model to include unrepressed foreign investors with limited risk-bearing capacity who can participate in both bond markets. I argue that financial repression is more effective when the government bond market is dominated by domestic investors whose demand is relatively inelastic. I call this the captive demand effect. On the other hand, the repression premium is harder to sustain when a large share of government debt is held by flighty foreign investors who can discipline the market price P_t^{Gov}.

Foreign investor problem. Let $B_{t+1}^{F,Gov}$ and $B_{t+1}^{F,Sup}$ denote the quantities of LC government and supranational bonds held by foreign investors at time t to be carried into $t+1$. Foreign investors do not receive private convenience services from EM LC bond holdings, as they do not operate inside the domestic financial system, and they are not subject to the financial repression constraint. In addition to the baseline liquidity frictions φ_t^{Gov} and φ_t^{Sup}, foreign investors also incur a quadratic portfolio management cost in each market, $(\psi_t^{Gov}/2)(B_{t+1}^{F,Gov})^2$ and $(\psi_t^{Sup}/2)(B_{t+1}^{F,Sup})^2$, where $\psi_t^{Gov}, \psi_t^{Sup} > 0$. This reduced-form cost can represent, for example, restrictive investment mandates or funding frictions that make it less attractive to take large positions in either asset. Let M_{t+1}^F be the foreign investors' stochastic discount factor in LC units,[9] and define

$$q_t^F \equiv \mathbb{E}_t\big[M_{t+1}^F\big], \qquad r_t^F \equiv -\log q_t^F. \tag{12}$$

Foreign investors choose their holdings to maximise the net payoff function

$$\begin{aligned}\max_{B_{t+1}^{F,Gov}, B_{t+1}^{F,Sup} \geq 0} \quad & \mathbb{E}_t\big[M_{t+1}^F(1-\delta_{t+1})\big] B_{t+1}^{F,Gov} + q_t^F\, B_{t+1}^{F,Sup} - \big(P_t^{Gov} + \varphi_t^{Gov}\big) B_{t+1}^{F,Gov} \\ & - \big(P_t^{Sup} + \varphi_t^{Sup}\big) B_{t+1}^{F,Sup} - \frac{\psi_t^{Gov}}{2}\big(B_{t+1}^{F,Gov}\big)^2 - \frac{\psi_t^{Sup}}{2}\big(B_{t+1}^{F,Sup}\big)^2.\end{aligned} \tag{13}$$

The two positions enter the problem separately, so the first-order conditions are

$$P_t^{Gov} + \varphi_t^{Gov} + \psi_t^{Gov} B_{t+1}^{F,Gov} = \mathbb{E}_t\big[M_{t+1}^F(1-\delta_{t+1})\big], \tag{14}$$

$$P_t^{Sup} + \varphi_t^{Sup} + \psi_t^{Sup} B_{t+1}^{F,Sup} = q_t^F. \tag{15}$$

Define foreign investors' fundamental valuations net of transaction costs as

$$V_t^{F,Gov} \equiv \mathbb{E}_t\big[M_{t+1}^F(1-\delta_{t+1})\big] - \varphi_t^{Gov}, \qquad V_t^{F,Sup} \equiv q_t^F - \varphi_t^{Sup}. \tag{16}$$

Then we obtain foreign demands that are linear in the respective market prices,

$$B_{t+1}^{F,Gov} = \frac{V_t^{F,Gov} - P_t^{Gov}}{\psi_t^{Gov}}, \qquad B_{t+1}^{F,Sup} = \frac{V_t^{F,Sup} - P_t^{Sup}}{\psi_t^{Sup}}. \tag{17}$$

Hence, positive ψ_t^{Gov} and ψ_t^{Sup} make foreign demand in each market finite and downward sloping.

Market clearing. Let $\bar{B}_t^{Gov}$ and $\bar{B}_t^{Sup}$ be the exogenously given total outstanding supplies of government and supranational bonds at time t; both may vary over time but are inelastic within the period. Market clearing requires

$$B_{t+1}^{D,Gov} + B_{t+1}^{F,Gov} = \bar{B}_t^{Gov}, \qquad B_{t+1}^{D,Sup} + B_{t+1}^{F,Sup} = \bar{B}_t^{Sup}, \tag{18}$$

where $B_{t+1}^{D,Gov}$ and $B_{t+1}^{D,Sup}$ are the domestic investor's holdings in equilibrium. Define the foreign and domestic ownership shares in the government bond market

$$s_t^F \equiv \frac{B_{t+1}^{F,Gov}}{\bar{B}_t^{Gov}}, \qquad s_t^D \equiv 1 - s_t^F. \tag{19}$$

[9]One can interpret M_{t+1}^F as the foreign discount factor after any currency hedging strategy has been applied.

Given the empirical patterns of bond holdings, I focus on the case in which all investor classes hold strictly positive amounts of both bonds, so that (6), (7), (14), and (15) all hold with equality.

The repression premium and sovereign investor base. Using (17) and (18)–(19), the equilibrium government bond price can be expressed as

$$P_t^{Gov} = V_t^{F,Gov} - \psi_t^{Gov}\, s_t^F\, \bar{B}_t^{Gov}. \tag{20}$$

From the domestic Euler equation (7), define the domestic investor's fundamental valuation of the government bond net of transaction costs,

$$V_t^{D,Gov} \equiv \mathbb{E}_t\big[\tilde{M}_{t+1}(1-\delta_{t+1})\big] + \chi_t - \varphi_t^{Gov}. \tag{21}$$

Then (7) implies $P_t^{Gov} = V_t^{D,Gov}/(1-\rho_t)$, or equivalently

$$\rho_t = 1 - \frac{V_t^{D,Gov}}{P_t^{Gov}}. \tag{22}$$

Imposing the Kuhn-Tucker condition $\rho_t \geq 0$ and combining (20) and (22) yields a mapping from the foreign ownership share to the financial repression premium:

$$\rho_t = \max\left\{0,\ 1 - \frac{V_t^{D,Gov}}{V_t^{F,Gov} - \psi_t^{Gov} s_t^F \bar{B}_t^{Gov}}\right\}. \tag{23}$$

Hence,

$$\rho_t > 0 \quad \Longleftrightarrow \quad s_t^F < \frac{V_t^{F,Gov} - V_t^{D,Gov}}{\psi_t^{Gov}\, \bar{B}_t^{Gov}}. \tag{24}$$

As such, there is a threshold foreign ownership rate below which the policy wedge is positive. Equation (24) formalises the idea that financial repression is more effective at reducing borrowing costs when the residual government bond supply that has to be absorbed by foreign investors is smaller.

Neither the policy parameter κ_t nor the wedge ρ_t is observable, so testing the model requires an observable counterpart. The ownership structure of the government bond market provides one. When the constraint binds, domestic holdings are set by regulation, $B_{t+1}^{D,Gov} = \kappa_t W_t / P_t^{Gov}$, and foreign investors absorb the remaining supply along their demand curve. Substituting the equilibrium price (20) into the binding constraint then yields an equation that ties the domestic ownership share to the policy parameter,

$$s_t^D\, \bar{B}_t^{Gov}\Big[V_t^{F,Gov} - \psi_t^{Gov}\big(1-s_t^D\big)\bar{B}_t^{Gov}\Big] = \kappa_t W_t, \tag{25}$$

whose left-hand side is strictly increasing in s_t^D. Holding $\big(V_t^{F,Gov}, \psi_t^{Gov}, W_t, \bar{B}_t^{Gov}\big)$ fixed, a tighter constraint implies a larger domestic share (see Appendix A). Conditional on a rich set of market controls, the domestic ownership share therefore tracks repression intensity, and I use it as the observable proxy for financial repression in the empirical analysis.

The supranational market. No constraint operates in the supranational market, so the domestic Euler equation (6) prices the bond at the domestic valuation $V_t^{D,Sup} \equiv \tilde{q}_t + \chi_t - \varphi_t^{Sup}$ with no repression wedge. Equating this with the foreign valuation condition (15) gives

$$V_t^{D,Sup} = V_t^{F,Sup} - \psi_t^{Sup} B_{t+1}^{F,Sup}, \tag{26}$$

which determines how the supranational stock is split across the two investor classes. Since χ_t is decreasing in domestic holdings and $\psi_t^{Sup} > 0$, the split is unique. In the government bond market the two classes' valuations are reconciled by the wedge ρ_t; in the supranational market they are reconciled by quantities alone. Equation (6) is a marginal condition and requires only that the domestic investor willingly holds *some* of the supranational bond stock. In equilibrium the price satisfies (6) and (15) simultaneously, so the pricing equations of Section 2.1 apply even when a large share of the supranational stock is held offshore.[10]

2.3 Separating currency and non-currency components

Now that ρ_t is linked to an observable proxy s_t^D, a natural question is whether the government yield alone is sufficient to empirically test the existence of a financial repression premium in EM LC government bonds. Based on the model, one cannot separately identify the financial repression premium ρ_t from government yields alone, because financial repression does not only impact ρ_t but the shadow risk-free rate $\tilde{r}_t$ in the economy as well. Note that

$$\tilde{r}_t = -\log\left(\mathbb{E}_t[\tilde{M}_{t+1}]\right) = -\log\left(\beta\,\mathbb{E}_t\left[\frac{u'(c_{t+1}) - \lambda_{t+1}\kappa_{t+1}}{u'(c_t)}\right]\right), \tag{27}$$

hence if the government increases the minimum-holdings requirement at time t, and the policy is expected to persist, the shadow risk-free rate at time t can increase. In that case, the overall direction of government bond prices will be ambiguous. Similarly, the χ_t term is unobserved and it moves with the equilibrium asset allocation. Therefore, to test the sensitivity of government borrowing costs to financial repression, both $\tilde{r}_t$ and χ_t in (10) must be accounted for; since neither is observed, neither can be controlled for in a regression. The rest of this section shows that the DBP is a useful device to account for these unobserved components.

Domestic bond premium Let $y_{j,t,\tau}^{Gov}$ and $y_{j,t,\tau}^{Sup}$ denote the zero-coupon yields at time t on a government bond and a AAA-rated supranational bond in currency j and tenor τ. The DBP is defined as

$$DBP_{j,t,\tau} \equiv y_{j,t,\tau}^{Gov} - y_{j,t,\tau}^{Sup}. \tag{28}$$

[10] If one investor class is absent from a market, that leg is priced by the other class alone and the DBP additionally includes a market segmentation wedge. To the extent that such segmentation reflects limited capital account openness, it can itself be read as financial repression, which operates in part through restrictions on capital flows; and because it is slow-moving, it can be absorbed by the currency–tenor and currency–tenor–year fixed effects in the empirical analysis.

Using the decomposition (11), we can expand the expression:

$$DBP_{j,t,\tau} = \underbrace{\left(\tilde{r}_{j,t,\tau} - \tilde{r}_{j,t,\tau}\right)}_{=0} + \underbrace{\left(I^{Gov}_{j,t,\tau} - I^{Sup}_{j,t,\tau}\right)}_{=I^{Gov}_{j,t,\tau}} + \underbrace{\left(L^{Gov}_{j,t,\tau} - L^{Sup}_{j,t,\tau}\right)}_{\equiv \widehat{L}^{GS}_{j,t,\tau}} + \underbrace{\left(R^{Gov}_{j,t,\tau} - R^{Sup}_{j,t,\tau}\right)}_{=-\rho_{j,t,\tau}}, \tag{29}$$

where the last term follows because the private convenience premium is common to the two bonds: $R^{Gov} - R^{Sup} = (-\chi/\tilde{q} - \rho) - (-\chi/\tilde{q}) = -\rho$. Crucially, since both bonds are in identical currency and tenor, the risk-free rates also cancel out. Dropping the tenor subscripts for ease of notation, one obtains

$$DBP_{j,t} \approx \underbrace{\frac{\mathbb{E}_t[\tilde{M}_{t+1}\delta_{j,t+1}]}{\tilde{q}_{j,t}}}_{I^{Gov}} + \underbrace{\frac{\varphi^{Gov}_{j,t} - \varphi^{Sup}_{j,t}}{\tilde{q}_{j,t}}}_{\widehat{L}^{GS}} - \rho_{j,t}. \tag{30}$$

Empirically, the default-risk and liquidity terms in (30) admit natural proxies (e.g., sovereign CDS spreads and bid-ask spreads), so the data can be used to estimate the DBP's sensitivity to these components. Conditional on those proxies, movements in the DBP within a currency and tenor trace movements in the financial repression wedge. Equation (30) thus motivates an empirical approach that (i) controls for I^{Gov} and $\widehat{L}^{GS}$ using observable proxies, (ii) includes currency-tenor fixed effects, which absorb persistent cross-sectional differences in the DBP, and (iii) evaluates the financial repression interpretation by relating the remaining variation to domestic investor concentration in government bond markets, which was shown to be related to the equilibrium level of ρ_t in Section 2.2.

In summary, the theoretical framework delivers clear sign predictions that can be tested empirically. Holding other components fixed: (i) increases in sovereign default risk raise $I^{Gov}_{j,t,\tau}$ and should be associated with a higher DBP; (ii) deteriorating supranational liquidity raises L^{Sup} and reduces $\widehat{L}^{GS}$, lowering the DBP; and (iii) more intense financial repression would raise $\rho_{j,t}$ and compress the DBP.[11]

3 Data

Supranational bond data. To estimate the DBP and assess its sensitivity to the risk premium components derived in Section 2, I construct a novel dataset of supranational zero-coupon yield curves for 11 EM currencies. These yield curves are built from security-level data covering secondary market prices and characteristics of bonds issued by eight major supranational institutions sourced from Bloomberg. The bond universe is restricted to fixed-rate coupon and zero-coupon securities; inflation-linked, floating-rate, callable, and putable bonds are excluded.

The choice of currencies was dictated by data availability, as the time-series estimation of

[11] An alternative measure of the non-currency component of EM LC government yields is the local currency credit spread (LCCS) of Du and Schreger (2016), defined as the spread between EM LC government bond yields and currency-hedged US Treasury yields. Because its two legs are in different currencies, the cancellations that deliver (30) no longer operate, and the LCCS instead retains the CIP basis from the currency hedge and the convenience yield on US Treasuries, both of which reflect global conditions rather than domestic policy. Appendix B derives the LCCS decomposition, shows when the CIP condition can fail, and compares the two measures empirically.

smooth and reliable yield curves requires overcoming a major challenge: having a sufficient number of secondary market quotes on each trading day. To address this challenge, I pool bonds across the following supranational institutions: the International Bank for Reconstruction and Development (IBRD), International Finance Corporation (IFC), European Bank for Reconstruction and Development (EBRD), European Investment Bank (EIB), Asian Development Bank (ADB), Kreditanstalt für Wiederaufbau (KfW), Inter-American Development Bank (IADB), and the African Development Bank (AfDB).[12] All eight maintain AAA long-term credit ratings throughout the sample period, ensuring that the resulting benchmark is homogeneous from a credit risk perspective. While no single issuer provides sufficient coverage across all currencies and maturities, pooling the eight institutions yields near-complete coverage of the supranational bond market in EM currencies.

Table 1: Supranational bond issuance by currency and issuer

	EIB	IBRD	EBRD	IFC	ADB	KfW	IADB	AfDB	**All**
TRY	23,725	6,931	19,044	7,455	7,216	6,263	1,538	1,826	**73,998**
ZAR	13,785	18,211	10,953	2,663	6,171	5,097	1,269	6,846	**64,997**
BRL	8,342	11,339	7,769	8,414	3,704	5,501	2,765	1,542	**49,374**
CNY	1,081	5,006	1,907	3,198	5,504	7,480	0	1,424	**25,601**
MXN	3,442	6,712	722	5,624	975	886	2,053	988	**21,402**
INR	737	5,829	5,364	3,056	1,462	32	3,882	326	**20,687**
PLN	14,625	1,129	697	7	940	802	0	0	**18,199**
RUB	2,262	3,236	3,577	2,737	468	691	48	455	**13,476**
IDR	2,234	1,415	5,040	131	158	133	3,051	229	**12,391**
HUF	2,059	97	241	292	263	386	0	0	**3,338**
COP	0	1,149	118	591	995	0	74	0	**2,927**
All	**72,293**	**61,055**	**55,432**	**34,168**	**27,856**	**27,271**	**14,679**	**13,637**	**306,392**

Notes: This table reports the aggregate amount issued by an issuer in a given currency. Units are in millions of US dollars. The sample contains 4,017 supranational bonds issued between 1997 and 2025 across 11 EM currencies. BRL = Brazilian real; CNY = Chinese renminbi; COP = Colombian peso; HUF = Hungarian forint; IDR = Indonesian rupiah; INR = Indian rupee; MXN = Mexican peso; PLN = Polish zloty; RUB = Russian rouble; TRY = Turkish lira; ZAR = South African rand. Sources: Bloomberg and author's calculations.

The final dataset comprises 4,017 supranational bonds across 11 EM currencies, covering issuance between 1997 and 2025. Table 1 details the issuance patterns by currency and institution. No single issuer dominates across all currencies, confirming that pooling the eight institutions is necessary to construct a sufficiently deep default-free benchmark curve in each currency.

Supranational yield curves. Since most supranational bonds are coupon-bearing instruments with varying maturities, computing the DBP requires extracting zero-coupon yields that are comparable to government benchmark rates at fixed tenors. I construct these supranational zero-coupon yield curves via a Dynamic Nelson-Siegel term-structure model (see Diebold and

[12]Strictly speaking, KfW is not a multilateral institution but a German government-owned development bank whose obligations carry an explicit statutory guarantee from the Federal Republic of Germany. The guarantee makes its bonds free of outright default risk. Because KfW shares the properties that matter for the analysis, namely AAA credit quality, offshore issuance in EM currencies, and independence from EM domestic financial regulation, I retain it in the pool and refer to all eight issuers as supranationals for brevity.

Li, 2006; Diebold et al., 2006). The original Nelson and Siegel (1987) model parsimoniously represents the term structure of interest rates through three latent factors interpreted as *level*, *slope*, *curvature*; this parsimony is crucial when only a limited number of supranational bonds trade on any given day. The Dynamic Nelson-Siegel model is well-suited to handle sparse, unbalanced bond price panels and maps naturally into a non-linear state-space system, which I estimate by an extended Kalman filter. The technical details of the estimation procedure are presented in Appendix D.

Government yield curves. Unlike supranational bonds in EM currencies, government bond yield curves are readily available from commercial data providers. For each currency, I obtain the government zero-coupon yields at 1, 3, 5, and 10-year tenors directly from Bloomberg. To ensure a complete time-series for the entire sample period, I impute missing observations using equivalent series from Refinitiv. Further details on data sources are presented in Appendix C.

4 Stylised facts

Table 2 presents summary statistics for government and supranational yields as well as the DBP for 11 EM currencies in my sample. Across currencies, supranational and government zero-coupon yields are broadly similar in levels, but the DBP widens as maturity lengthens. Governments typically yield below supranationals at shorter maturities (median DBP of -28 bp at the one-year tenor), but above them at longer maturities (median $+32$ bp at the 10-year tenor). Standard deviations range from 1.7 to 3.3 percentage points, and at the short-end of the yield curve the DBP can vary from $+5$ to -41 percentage points. Such extreme variation occurs primarily due to episodes of severe market stress in Russia and Turkey, where increases in financial repression and policy-induced market segmentation contributed to highly negative DBPs (Section 5.5 examines these episodes in detail).

Figure 2 plots the cross-currency averages of the EM LC government–US Treasury spread and the DBP over time. Overall, three patterns emerge. First, currency risk, reflected by the gap between the two lines, accounts for most of the variation in EM sovereign credit spreads throughout the sample period. The persistence of this pattern underscores the fact that investors primarily demand compensation for currency depreciation risk rather than nominal repayment risk when holding EM LC government debt.

Second, the DBP exhibits a secular decline. At five-year tenor, it falls from around 150 bp in 2012 to near zero as of 2025. This compression coincided with several positive developments in emerging markets: improvements in monetary and fiscal policy conduct, the inclusion of EM LC bonds in global fixed-income indices such as the J.P. Morgan GBI-EM, and the rapid growth of domestic institutional investor bases. The declining trend suggests that investors may increasingly be viewing EM LC sovereign debt as a legitimate asset class with improving fundamentals (e.g., see Arslanalp and Tsuda, 2014; Du and Schreger, 2022b).

Third, the DBP exhibits stability during stress episodes such as the GFC, Covid-19 crisis, and the post-pandemic surge in inflation. While EM LC yield spreads over US Treasuries widen dramatically during these periods, the DBP remains below two percentage points or even turns

Table 2: Summary statistics of EM LC zero-coupon yields and the DBP

		Tenor			
		1-year	3-year	5-year	10-year
Median	Supranational	6.853	6.888	7.014	7.250
	Government	6.590	6.947	7.263	7.644
	DBP	−0.284	−0.003	0.123	0.321
Std. dev.	Supranational	6.700	5.326	4.500	3.662
	Government	5.348	4.538	4.021	3.354
	DBP	3.260	2.555	2.118	1.723
Minimum	Supranational	0.101	0.110	0.433	−0.103
	Government	−0.024	0.107	0.429	1.178
	DBP	−40.500	−33.006	−26.127	−19.630
Maximum	Supranational	57.261	49.070	41.297	32.168
	Government	50.621	43.018	34.753	25.784
	DBP	4.911	5.529	6.133	7.836

Notes: This table reports the median, standard deviation, minimum, and maximum values for the zero-coupon yields and the DBP across 11 EM currencies throughout the sample period (2007-2025). Units for yields are in percent and DBP is in percentage points. Sources: Bloomberg, Refinitiv, and author's calculations.

negative. These patterns are particularly pronounced at shorter maturities and are robust to excluding the outliers Russia and Turkey. Why would the DBP not rise when the EM LC government–US Treasury spread widens? Several explanations merit consideration.

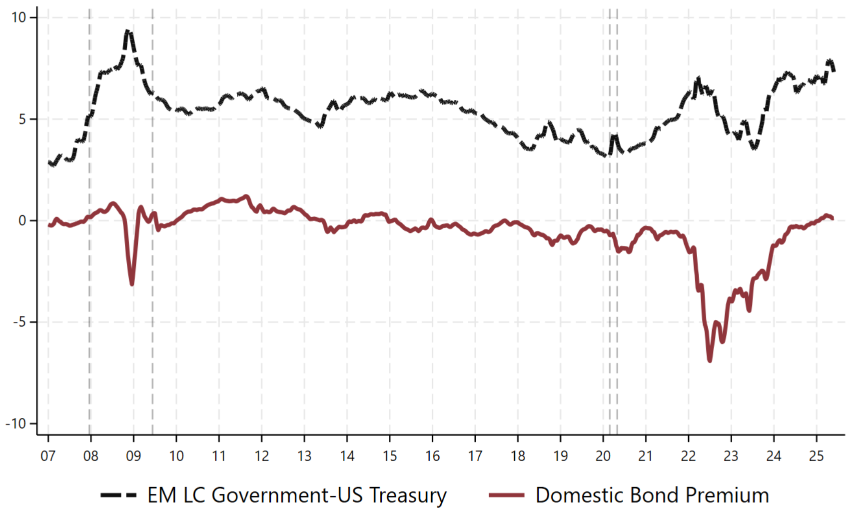
One explanation is that much of the default risk on LC government bonds manifests through inflation and the resulting currency depreciation rather than outright default (e.g., see Reinhart and Rogoff, 2011). Since both government and supranational bonds face identical FX exposure, the DBP does not capture this important channel of sovereign risk by construction. This could explain the relative stability of DBPs.

Alternatively, central banks typically adopt a counter-cyclical monetary policy stance, lowering rates to stimulate economic activity and reduce borrowing costs during crises (De Leo et al., 2022). However, this alone is unlikely to explain the DBP compression, because both government and supranational bonds are denominated in local currency. Therefore, unless the investor composition for the two bond types is different, rate cuts should affect both bonds similarly, leaving the DBP broadly unchanged.

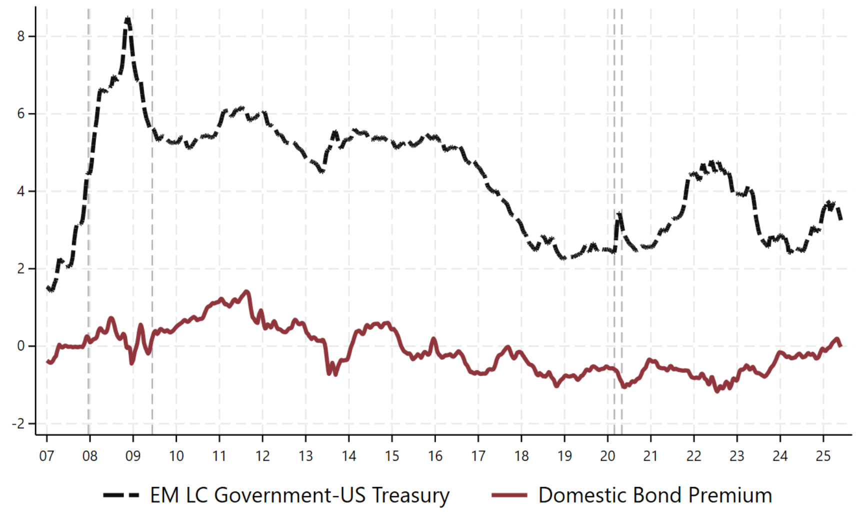
A third possibility is that secondary market liquidity for supranational bonds could deteriorate more during crises, compressing the DBP through higher supranational yields. This channel likely plays some role: supranational liquidity deteriorates when sovereign credit risk rises, and this illiquidity is associated with lower DBPs. However, two considerations suggest this is only a partial explanation. First, as I show in Section 5, supranational illiquidity accounts for a modest amount of DBP variation once other factors are controlled for. Second, government bond liquidity is likely to deteriorate during crises, limiting any differential effect on the spread.

Crucially, government and supranational bonds may exhibit different price dynamics, especially during stress episodes, due to differences in their investor bases. While supranational

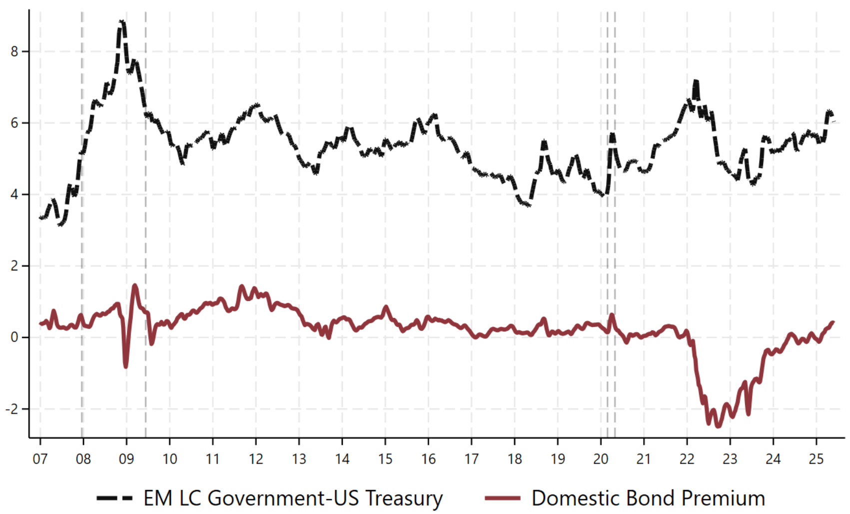
Figure 2: Cross-country averages of the EM LC government–US Treasury spread and the DBP



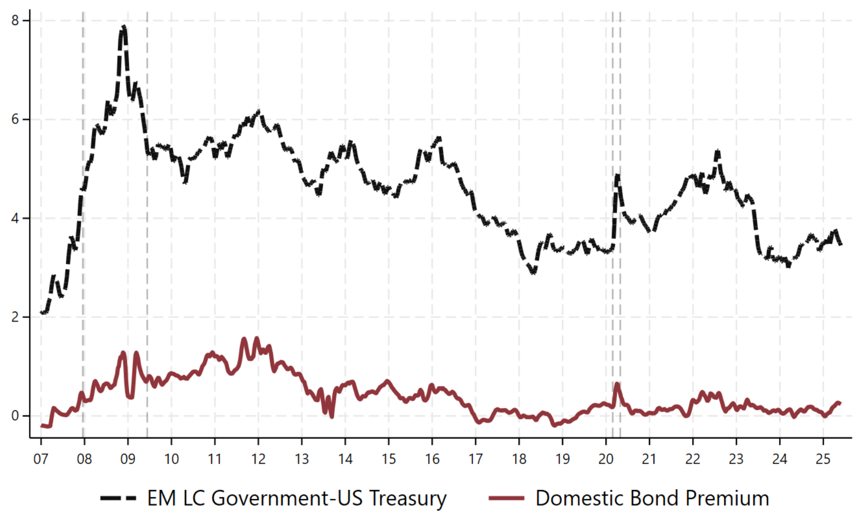
(a) One-year tenor (full sample)



(b) One-year tenor (ex-Russia & Turkey)



(c) Five-year tenor (full sample)



(d) Five-year tenor (ex-Russia & Turkey)

Notes: This figure shows the cross-country, four-week rolling averages of the EM LC government–US Treasury spread (black) and the DBP (red) at weekly frequency. Units are in percentage points. The full sample comprises BRL, CNY, COP, HUF, IDR, INR, MXN, PLN, RUB, TRY, and ZAR; the ex-Russia & Turkey panels exclude RUB and TRY. Sources: Bloomberg, Refinitiv, and author's calculations.

bond yields spike during stress periods, government bonds maintain stable or even declining yields due to strong domestic demand. This pattern may reflect financial repression and policy-induced market segmentation. By compelling domestic institutions to hold LC government debt regardless of risk-return considerations, governments can dampen the impact of external shocks on their borrowing costs. In this case, offshore supranational bonds give us a proxy for 'what would have happened' if the domestic bond market was not constrained.

The ability of EM governments to borrow near or below default-free rates raises a critical question for debt sustainability analysis: does this reflect genuine improvements in creditworthiness, or captive demand created through regulatory pressure on domestic institutions? The next section disentangles these explanations by estimating the sensitivity of DBP to sovereign default risk and financial repression, controlling for supranational illiquidity.

5 What explains the domestic bond premium?

5.1 Global versus local factors

If financial repression is an important factor in EM LC government bond pricing, one would expect the DBP variation to be predominantly country-specific, as repressive policies respond to domestic fiscal pressures. To test this prediction, I decompose the overall DBP variance into components attributable to persistent country characteristics, common global shocks, and idiosyncratic local factors.[13]

Table 3: Variance decomposition of the DBP

Tenor	**Between-variance**	**Within-variance**	
		Local factors	Global factors
1-year	11.73%	69.32%	18.95%
3-year	11.59%	72.22%	16.19%
5-year	14.17%	72.84%	12.99%
10-year	20.61%	68.68%	10.71%

Notes: This table reports the variance decomposition of the DBP panel during the sample period (2007–2025). Data aggregated to monthly frequency prior to decomposition. See Appendix E for further details on the methodology. Sources: Bloomberg, Refinitiv, and author's calculations.

The decompositions in Table 3 show that, at each maturity length, around 70% of the overall variance in DBP can be explained by idiosyncratic and time-varying (i.e., local) factors. Meanwhile, common shocks that hit all markets simultaneously (global factors) account for only 10-19% of the variance. The dominance of idiosyncratic factors contrasts with Longstaff et al. (2011), who document a large common component in sovereign CDS spreads across countries.

The stark difference reflects DBP's ability to exclude currency risk premiums, which are highly correlated across EMs. Indeed, a principal component (PC) analysis on EM exchange rates in my sample between 2007 and 2025 vis-à-vis the US dollar reveals that the first PC

[13] The decomposition serves an additional purpose: by revealing the importance of global factors, it indicates whether time fixed effects should be included in panel regressions. Appendix E provides further details on the methodology.

explains 81% of the total variance in exchange rate movements. This large common component suggests limited risk-sharing across EM currencies.

Furthermore, persistent cross-country differences (between variance) explain at most 21% of the DBP variance, suggesting that time-varying factors rather than fixed country characteristics account for most of the variation in government borrowing costs relative to default-free benchmarks. Overall, these patterns are consistent with local macro-financial conditions or domestic policies, including financial repression, driving the yield differentials.[14]

The percentages in Table 3 also provide upper bounds on what different sets of covariates can achieve: global time-varying factors *alone* cannot explain more than 19% of the DBP variation, and fixed cross-sectional controls are capped at around 20%. As such, my empirical strategy relies on currency-tenor fixed effects and time-varying controls, with an emphasis on understanding the contribution of time-varying variables that proxy for the DBP components.

5.2 Testing the equilibrium predictions of the framework

This section tests the predictions of the framework presented in Section 2 by assessing the sensitivity of DBP to each risk premium component. I map the model-implied decomposition (30) to a panel regression that controls for sovereign default risk with CDS spreads[15], the relative liquidity premium $\widehat{L}^{GS}_{j,t,\tau}$ with government and supranational bond bid-ask spreads, and absorbs persistent cross-sectional differences in the DBP with currency–tenor fixed effects.

The theoretical framework predicts that the financial repression wedge $\rho_{j,t}$ increases with the domestic ownership share of government bonds, as a higher concentration of captive domestic investors makes repression more effective keeping all else constant. I therefore use the domestic ownership share as a proxy for the intensity of financial repression.

However, financial repression need not operate solely through captive domestic demand. Inflationary monetary policy accompanied by regulatory policies that trap domestic savers can also compress borrowing costs, so lower real policy rates often go hand-in-hand with financial repression (Reinhart and Sbrancia, 2015; Kose et al., 2022). I therefore include the real policy rate as an additional control in the baseline regression. I measure it using central bank policy rates rather than government bond yields for two reasons: first, government bond yields are an input to the DBP, so using them would introduce mechanical correlation; second, central bank rates better reflect the intended monetary policy stance, whereas government bond yields may also incorporate factors such as domestic financial intermediaries' external funding costs (e.g., De Leo et al., 2022). The resulting specification is:

$$\begin{aligned} DBP_{j,t,\tau} = \alpha + \beta_1 CDS_{j,t} + \beta_2 BidAsk^{Gov}_{j,t,\tau} + \beta_3 BidAsk^{Sup}_{j,t,\tau} + \beta_4 Dom.\ Holdings_{j,t-1} \\ + \beta_5 Real\ Policy\ Rate_{j,t-1} + \gamma' X_{j,t} + \phi_{j,\tau} + \nu_{j,t,\tau}, \end{aligned} \tag{31}$$

[14] However, the decomposition should be interpreted conservatively, as global shocks may interact with local economic conditions and time-varying factors can eventually influence persistent country characteristics.

[15] I use the five-year sovereign CDS spread, the most liquid contract with the broadest cross-country coverage. Section 5.3 checks robustness to using tenor-matched CDS-implied default probabilities. Crucially, although CDS contracts predominantly reference foreign currency obligations, the mismatch with LC bonds is limited. The risk channel specific to LC debt, de facto default through inflation and currency depreciation, affects both legs of the DBP symmetrically and nets out of the spread by construction. The remaining component, outright default risk, is similar across denominations; the average gap between EM sovereigns' local and foreign currency credit ratings, as wide as three notches in the mid-1990s, has since closed almost entirely (Amstad et al., 2020).

where $Dom.\ Holdings_{j,t-1}$ is the lagged level of the domestic ownership share of LC government bonds. $Real\ Policy\ Rate_{j,t-1}$ captures the lagged real policy rate. Based on the decomposition (30), I expect $\beta_4 < 0$ and $\beta_5 > 0$: higher domestic ownership of LC government bonds and lower real policy rates should be followed by a smaller DBP. Moreover, higher sovereign default risk should be associated with a higher ($\beta_1 > 0$) and lower supranational liquidity with a smaller DBP ($\beta_3 < 0$).

The estimation sample includes all 11 EM currencies but excludes Russian rouble observations from February 2022 and Turkish lira observations from May 2022 onward, to ensure that the baseline estimates are not driven by extreme repression episodes in which the DBP reached -30 percentage points or below. These episodes are examined separately as case studies in Section 5.5. All regressors are standardised at the currency–tenor level to facilitate comparison across variables with different units; the resulting coefficients measure the change in DBP (in percentage points) associated with a one-standard-deviation increase in each regressor. Following Du and Schreger (2016) and Dao and Gourinchas (2026), I estimate the regression equation using an OLS fixed-effects estimator with Driscoll and Kraay (1998) heteroskedasticity and autocorrelation-adjusted standard errors.[16]

Table 4 reports the results. The financial repression proxies are statistically significant and stable across all specifications. The domestic holdings coefficient is negative and significant at all maturities: it ranges from -0.15 to -0.23 at the one-year tenor and from -0.19 to -0.26 at longer maturities, indicating that a one-standard-deviation increase in the domestic ownership share is associated with a 15–26 basis point decline in the DBP. The real policy rate coefficient is positive and significant throughout, ranging from 0.30 to 0.32 at the one-year tenor and from 0.17 to 0.19 at longer maturities. The stability of the domestic holdings coefficient when the real policy rate is added suggests that these two proxies capture distinct but related channels of government policy: generating inelastic domestic demand supplemented with accommodative monetary policy.

Sovereign default risk exhibits a maturity divide. At longer maturities, the CDS coefficient is positive, large, and significant across all specifications (0.22–0.24), indicating that a one-standard-deviation increase in sovereign default risk is associated with a 22–24 basis point rise in DBP. At the one-year tenor, by contrast, the CDS coefficient is statistically insignificant throughout, consistent with the low probability of outright nominal default over a one-year horizon for most EMs in the sample.

Supranational illiquidity retains its expected negative sign and is significant across all specifications at both maturities, with a stronger effect at the one-year tenor (-0.23 to -0.39) than at longer maturities (-0.13 to -0.18). Government bond illiquidity is insignificant at short maturities; at longer maturities it is weakly positive in some specifications but attenuates in the full specification, suggesting it partly captures omitted global factors.

The coefficients also reveal a term structure of financial repression, though the pattern differs across channels. The real policy rate sensitivity is roughly twice as large at the one-year tenor as at longer maturities, consistent with central bank policy rates anchoring the short

[16]These standard errors are more appropriate than currency-level clustering when there are relatively few currencies (small N) and a long sample period (large T).

Table 4: Domestic bond premium, sovereign default risk, and financial repression

	$DBP_{j,t,\tau}$							
	1−year				> 1−year			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$CDS_{j,t}$	−0.016	−0.039	−0.040	−0.047	0.239***	0.234***	0.233***	0.220***
	(0.127)	(0.129)	(0.140)	(0.127)	(0.037)	(0.038)	(0.039)	(0.049)
$BidAsk^{Gov}_{j,t,\tau}$	−0.054	−0.016	−0.053	−0.065	0.025	0.053**	0.073**	0.027
	(0.054)	(0.054)	(0.050)	(0.043)	(0.027)	(0.026)	(0.031)	(0.030)
$BidAsk^{Sup}_{j,t,\tau}$	−0.393***	−0.380***	−0.336***	−0.234***	−0.138***	−0.127***	−0.178***	−0.158***
	(0.085)	(0.081)	(0.081)	(0.069)	(0.046)	(0.044)	(0.049)	(0.051)
$Dom.\ Holdings_{j,t-1}$	−0.234***	−0.190***	−0.148**	−0.175**	−0.250***	−0.224***	−0.262***	−0.192***
	(0.085)	(0.070)	(0.062)	(0.070)	(0.058)	(0.052)	(0.059)	(0.055)
$Real\ Policy\ Rate_{j,t-1}$		0.314***	0.301***	0.320***		0.168***	0.172***	0.186***
		(0.078)	(0.073)	(0.077)		(0.034)	(0.036)	(0.047)
Constant	−0.355***	−0.338***	−0.304***	−0.231***	0.258***	0.273***	0.236***	0.236***
	(0.101)	(0.087)	(0.076)	(0.058)	(0.040)	(0.040)	(0.050)	(0.048)
Domestic controls	No	No	Yes	Yes	No	No	Yes	Yes
Global controls	No	No	No	Yes	No	No	No	Yes
Observations	1,503	1,490	1,483	1,483	4,664	4,636	4,613	4,613
Within-R^2	0.112	0.169	0.194	0.304	0.100	0.123	0.150	0.184

Notes: This table reports panel regression results of the DBP on proxies for default risk, illiquidity, and financial repression consistent with the model in Section 2. Estimation sample excludes RUB from February 2022 and TRY from May 2022 onward. Data in monthly frequency. All regressors are standardised at the currency–tenor level. $Dom.\ Holdings_{j,t-1}$ is the lagged level of the domestic ownership share of LC government bonds. $BidAsk^{Gov}_{j,t,\tau}$ and $BidAsk^{Sup}_{j,t,\tau}$ are bid–ask spreads for government and supranational bonds. Domestic controls: Debt/GDP, FX volatility, domestic stock index returns. Global controls: VIX, broad dollar index (EM), US Treasury yields. All specifications include currency–tenor fixed effects. Driscoll and Kraay (1998) heteroskedasticity and autocorrelation robust standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: Bloomberg, Refinitiv, BIS, IMF, and author's calculations.

end of the yield curve more strongly than long-term rates, which average over expected future policy stances. The domestic holdings coefficient, by contrast, does not exhibit a clear maturity gradient and is, if anything, slightly larger at longer maturities. The remainder of this section examines the robustness of these findings.

5.3 Robustness and alternative specifications

A potential concern with the baseline specification is that the persistent components absorbed by the currency–tenor fixed effects, including any residual issuer-specific convenience differential or market segmentation, may not remain constant over time, as sustained policy interventions gradually reshape market structure and investor behaviour. If these components drift, currency–tenor fixed effects may be insufficient and the independent variables could spuriously capture this slow-moving variation.

To address this concern, I replace currency–tenor fixed effects with currency-tenor-year fixed effects, which allow these persistent components to shift each year. This absorbs all common annual variation within each currency–tenor pair, including any slow-moving drift in these components, so that the financial repression proxies can only be significant if they explain within-year, within-currency-tenor movements in the DBP due to discrete changes in government policy.

Table 5 reports the results. Columns (1) and (3) use the five-year CDS spread as the default risk proxy, while columns (2) and (4) replace it with tenor-matched CDS-implied default probabilities as a robustness check. Both specifications augment the baseline with an interaction between the default risk measure and the lagged domestic ownership share, which tests whether the sensitivity of the DBP to default risk depends on the composition of the (pre-determined) sovereign investor base.

The domestic holdings coefficient remains negative and significant: -0.35 at the one-year tenor and -0.22 at longer maturities in the CDS specification, with comparable magnitudes when tenor-matched default probabilities are used instead. The real policy rate retains its expected positive sign, though it attenuates at longer maturities once year-level variation is absorbed.

The interaction between default risk and lagged domestic holdings is negative and significant across specifications, consistent with the model's prediction that captive domestic demand insulates government borrowing costs from deteriorating creditworthiness. At mean domestic ownership, a one-standard-deviation increase in CDS spreads is associated with an increase in the long-end DBP by 32 basis points; at one standard deviation above the mean, this number falls to approximately 19 basis points. Taken together, the survival of the financial repression proxies under this demanding specification and the significant interaction between default risk and domestic ownership reinforce the interpretation that DBP helps track policy-driven distortions in government borrowing costs.

A remaining concern is that the domestic ownership share may partly reflect voluntary home bias or investor portfolio choices that move endogenously with yields. To mitigate this concern, Table 6 replaces it with the lagged average reserve requirement ratio from Federico et al. (2014). Higher reserve requirements can increase demand for government debt through both

Table 5: Does captive demand insulate government borrowing costs from default risk pricing?

	$DBP_{j,t,\tau}$			
	1-year		> 1-year	
	(1)	(2)	(3)	(4)
$CDS_{j,t}$	−0.075		0.322***	
	(0.099)		(0.105)	
$CDS_{j,t} \times Dom.\ Holdings_{j,t-1}$	−0.133*		−0.129***	
	(0.078)		(0.026)	
$Default\ Prob._{j,t,\tau}$		−0.060		0.148***
		(0.064)		(0.051)
$Default\ Prob._{j,t,\tau} \times Dom.\ Holdings_{j,t-1}$		−0.136**		−0.111***
		(0.059)		(0.034)
$BidAsk^{Gov}_{j,t,\tau}$	−0.044	−0.035	0.034	0.009
	(0.028)	(0.026)	(0.025)	(0.023)
$BidAsk^{Sup}_{j,t,\tau}$	−0.077*	−0.061	−0.028	−0.024
	(0.046)	(0.047)	(0.024)	(0.025)
$Dom.\ Holdings_{j,t-1}$	−0.348***	−0.299*	−0.224***	−0.198***
	(0.129)	(0.171)	(0.074)	(0.071)
$Real\ Policy\ Rate_{j,t-1}$	0.120**	0.135**	0.055	0.062*
	(0.057)	(0.052)	(0.034)	(0.032)
Constant	−0.942***	−0.485***	0.147***	−0.007
	(0.201)	(0.136)	(0.036)	(0.049)
Domestic controls	Yes	Yes	Yes	Yes
Global controls	Yes	Yes	Yes	Yes
Currency×tenor×year FE	Yes	Yes	Yes	Yes
Observations	1,483	1,240	4,613	4,159
Within-R^2	0.801	0.796	0.746	0.798

Notes: This table reports panel regression results of the DBP on proxies for default risk, illiquidity, and financial repression. The interaction between lagged domestic holdings and default risk is tested to assess whether captive domestic demand insulates government borrowing costs from deteriorating creditworthiness. Sample excludes RUB observations from February 2022 and TRY observations from May 2022 onward. All regressors are standardised at the currency–tenor level. $Default\ Prob._{j,t,\tau}$ is the tenor-matched default probability implied by the sovereign CDS. Domestic controls: Debt/GDP, FX volatility, domestic stock index returns. Global controls: VIX, broad dollar index (EM), US Treasury yield. Driscoll and Kraay (1998) heteroskedasticity and autocorrelation robust standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: Bloomberg, Refinitiv, BIS, IMF, and author's calculations.

direct and indirect channels. In some institutional settings, reserve requirements can be satisfied partly through holdings of government securities. More generally, even when required reserves must be held as balances at the central bank, higher requirements tighten banks' balance-sheet constraints and reduce the funds available for loans and other risky securities. In the remaining allocation, LC government bonds often receive preferred regulatory treatment; they carry low risk weights, qualify as high-quality liquid assets, and can often be pledged in central-bank liquidity operations, including repo auctions. Banks therefore tilt residual portfolios toward sovereign debt, raising captive demand. The lagged reserve ratio therefore provides a cleaner

Table 6: Reserve requirements and the domestic bond premium

	$DBP_{j,t,\tau}$			
	1−year		> 1−year	
	(1)	(2)	(3)	(4)
$Default\ Prob._{j,t,\tau}$	0.053	0.089	0.170**	0.216**
	(0.106)	(0.165)	(0.065)	(0.106)
$Reserve\ Requirement_{j,t-1}$	−0.229**	−0.297***	−0.240***	−0.291***
	(0.088)	(0.092)	(0.055)	(0.066)
$Real\ Policy\ Rate_{j,t-1}$	0.260**	0.227**	0.311***	0.328***
	(0.108)	(0.113)	(0.071)	(0.076)
Constant	−0.118	1.341***	0.402***	0.832**
	(0.117)	(0.138)	(0.066)	(0.395)
Liquidity controls	Yes	Yes	Yes	Yes
Domestic controls	Yes	Yes	Yes	Yes
Global controls	Yes	No	Yes	No
Year FE	No	Yes	No	Yes
Observations	275	275	774	774
Within-R^2	0.447	0.575	0.301	0.389

Notes: This table reports panel regression results of the DBP on proxies for default risk and financial repression. In contrast to Table 4, I use reserve requirements as an alternative variable to capture regulatory pressure on banks' balance sheets. Data in quarterly frequency. All regressors are standardised at the currency–tenor level. $Reserve\ Requirement_{j,t-1}$ is the lag of average reserve requirement ratio from Federico et al. (2014) updated through 2021Q3. $Default\ Prob._{j,t,\tau}$ is the tenor-matched CDS-implied default probability. Liquidity controls: gov and supra bid-ask spreads. Domestic controls: Debt/GDP, FX volatility, domestic stock index returns. Global controls: VIX, broad dollar index (EM), US Treasury yield. Data at quarterly frequency. All specifications include currency–tenor fixed effects. Driscoll and Kraay (1998) heteroskedasticity and autocorrelation robust standard errors in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sources: Bloomberg, Refinitiv, BIS, IMF, Federico et al. (2014), and author's calculations.

proxy for regulatory pressure on banks' balance sheets, mitigating concerns that the estimated relation simply reflects voluntary home bias or portfolio shifts in response to current bond yields. Nonetheless, the timing of reserve requirement changes may be endogenous to the business cycle. Therefore, to the extent that these requirements respond to the same economic conditions that affect the DBP, the empirical results are still interpreted as correlational.

The next section provides a complementary high-frequency test using discrete macroprudential policy events, and Section 5.5 presents two case studies in which financial repression policies decoupled government borrowing costs from creditworthiness.

5.4 Macroprudential policies and government borrowing costs

The panel regressions in Sections 5.2 and 5.3 document that the DBP comoves with proxies for financial repression, but they do not show whether the DBP responds at high frequency to policy actions that are predicted to shift captive demand for government debt. This section provides a validation exercise.

To do so, I construct a daily event panel from the narrative entries in the IMF Integrated Macroprudential Policy Database (iMaPP) (Alam et al., 2019), which records macroprudential policy actions across 134 countries through end-2024. From the detailed policy descriptions underlying this dataset, I extract day-precise announcement or effective dates when available and use a large language model (Anthropic Claude Sonnet 4.6) to classify each action. A structured system prompt requires the model to return pre-specified fields for each narrative entry: the announcement and effective dates, the focal policy action, whether the action is relevant to government bond demand, the predicted direction of the effect, and the transmission channel (Appendix F describes dataset construction in more detail). With this dataset, I estimate the conditional dynamic associations between macroprudential policies and the DBP using daily panel local projections (Jordà, 2005).

Many of the policy actions in my database involve standard macroprudential tools such as reserve requirements, liquidity coverage ratios, or FX restrictions, and need not be designed to extract resources from the financial sector or reduce sovereign funding costs. Yet even when macroprudential policies are motivated by financial-stability concerns, they can affect sovereign borrowing conditions by reshaping financial institutions' balance sheet constraints. When sovereign securities are the most liquid LC asset eligible to satisfy regulatory requirements, captive demand for government debt may increase even without an explicit fiscal objective.

This mechanism connects closely to the theory laid out in Section 2. A demand-increasing event is represented in the model as a rise in κ_t, which raises the equilibrium repression wedge ρ_t when the constraint binds. Reis (2021b) offers a supporting structural account in which macroprudential regulation is modelled as the share of government bonds that banks must hold, and tightening that requirement mechanically raises bond prices through the same captive-demand channel. Holding the other components of decomposition (30) fixed, both formulations deliver the same prediction: the DBP should fall following demand-increasing (tightening) events and rise following demand-decreasing (loosening) events. Macroprudential actions classified as not relevant for government bond demand, such as borrower-side loan restrictions, should leave the DBP unchanged; these neutral events therefore serve as a placebo.

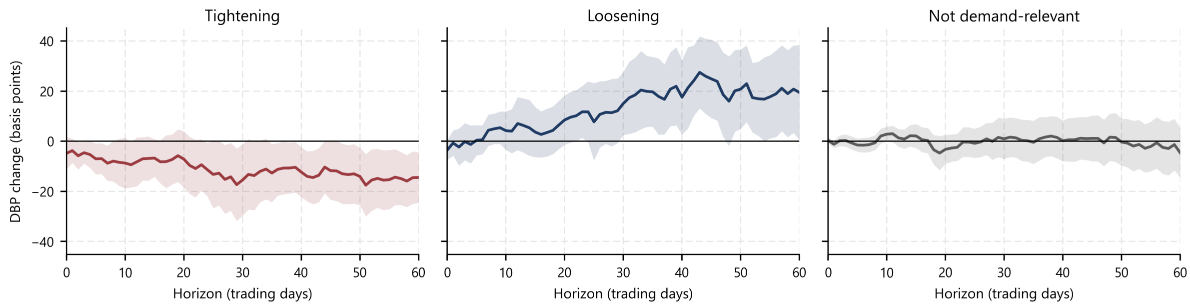
To test these predictions, let $U_{j,t}$, $D_{j,t}$, and $N_{j,t}$ indicate demand-increasing, demand-decreasing, and demand-neutral macroprudential policy dates, respectively. I measure the response as the cumulative change in the DBP from the trading day before the event to horizon h. For each currency–tenor pair (j, τ) and horizon $h \in \{0, 1, \dots, 60\}$ trading days, I estimate

$$\begin{aligned} DBP_{j,\tau,t+h} - DBP_{j,\tau,t-1} = {} & \beta_h^U U_{j,t} + \beta_h^D D_{j,t} + \beta_h^N N_{j,t} + \pi_h \Delta DBP_{j,\tau,t-1} \\ & + \theta_h^U U_{j,t-1} + \theta_h^D D_{j,t-1} + \theta_h^N N_{j,t-1} \\ & + \Gamma_h' X_{j,\tau,t-1} + \alpha_{j,\tau,y(t)} + \delta_t + \varepsilon_{j,\tau,t,h}, \end{aligned} \tag{32}$$

where $X_{j,\tau,t-1}$ is a vector of predetermined controls (the tenor-matched CDS-implied default probability and its 20-trading-day change, FX volatility, equity returns, the policy rate, and government and supranational bid–ask spreads), $\alpha_{j,\tau,y(t)}$ are currency–tenor–year fixed effects, and δ_t is a date fixed effect. The lagged change $\Delta DBP_{j,\tau,t-1}$ controls for short-run momentum in the premium. The fixed-effect structure absorbs both slow-moving country-specific drifts (analogous to the currency–tenor–year specification of Section 5.3) and global daily shocks.

Figure 3 plots the estimated response paths (i.e., the coefficients β_h^U, β_h^D, and β_h^N). On impact, the DBP is 7 basis points lower following a tightening event; the response builds gradually and reaches a trough of −20 basis points around 50 trading days after the event. Loosening events display a mirror-image pattern: a 15–25 basis point widening of the DBP over similar horizons. Both responses are statistically significant through most of the post-event window. Importantly, the estimated responses to neutral events hover around zero and are statistically insignificant at every horizon. The results are therefore consistent with the model's sign predictions.

Figure 3: DBP response following macroprudential policy events



Notes: This figure plots the estimated cumulative response of the DBP over 60 trading days following tightening, loosening, and demand-neutral macroprudential policy events. Estimates use Kalman-filtered but unsmoothed supranational yield series to avoid look-ahead bias in local projections. Tightening events (U, left panel) are policy actions predicted to increase demand for government bonds; loosening events (D, centre panel) are actions predicted to decrease demand; neutral events (N, right panel) are macroprudential actions classified as not relevant for government bond demand. The dependent variable is $DBP_{j,\tau,t+h} - DBP_{j,\tau,t-1}$ for $h = 0, \ldots, 60$. Standard errors are Driscoll and Kraay (1998) with Bartlett kernel and horizon-dependent bandwidth $\max\{m(T), h+1\}$, where $m(T) = \lfloor 4(T/100)^{2/9} \rfloor$. Shaded regions are 90% confidence bands. Sources: Bloomberg, Refinitiv, BIS, IMF, and author's calculations.

However, these estimates should not be interpreted as responses to exogenous financial repression shocks, as the timing of macroprudential actions is likely to depend on prevailing macroeconomic conditions. The relevant test is therefore more modest: whether the DBP moves at high frequency, and in the predicted direction, around policy actions that should increase or decrease captive demand for government debt.

The compression after demand-increasing events and the widening after demand-decreasing events support the interpretation of the DBP as a market-based measure that tracks policy-induced changes in domestic demand for government bonds. Moreover, the absence of a response to neutral events provides additional support for this interpretation. Finally, the pre-event diagnostic in Appendix F reduces the concern that these responses simply continue pre-existing DBP dynamics. Pre-event movements are statistically indistinguishable from zero and, for tightening events, the point estimates, if anything, drift in the opposite direction. The post-event responses in Figure 3 therefore reflect movement that begins at the event date rather than a continuation of pre-trends or anticipation effects. I now turn to two case studies where financial repression mechanisms are especially visible.

5.5 Case studies

5.5.1 Turkey

Figure 1 in the introduction plots Turkish lira-denominated five-year government and supranational yields over the full sample. After the Global Financial Crisis and before the 2018

Turkish lira crisis, the five-year sovereign yield tracked the supranational one closely, reflecting a near-zero DBP.

Between 2019 and 2022, government yields gradually diverged below supranational yields, with the DBP turning negative and reaching approximately -5 percentage points by early 2022. This period was characterised by a sequence of policies designed to reduce dollarisation and raise domestic demand for LC government bonds: raising required reserve ratios for foreign currency deposits, imposing limits on banks' swap transactions with foreign counterparties, and applying moral suasion to reduce foreign currency lending. These measures pushed domestic institutions toward lira assets, with government bonds being the primary option. In 2021, authorities introduced exchange rate-protected lira deposit accounts, which put significant strain on government finances.[17] Meanwhile, the central bank faced increasing political pressure; between 2019 and 2023, the governor was replaced four times by the president and policy rates were instructed to be kept low despite rising inflation (see Gürkaynak et al., 2023). During the same period, foreign participation in the LC government bond market declined by 15 percentage points, increasing the concentration of domestic holders.

These measures were followed by the most salient financial repression policy: the June 2022 announcement of the Securities Maintenance Practice (SMP).[18] This regulatory measure mandated banks to hold long-term LC government bonds in blocked accounts at the central bank, with requirements that linked their operational flexibility, including loan growth and interest rates, to their holdings of these securities (Celebi et al., 2025). Between mid-2022 and mid-2023, five-year nominal government yields declined by 20 percentage points despite 80% annual inflation and multiple credit rating downgrades. Following the gradual easing of the SMP requirements and monetary tightening after June 2023, yields reversed sharply, with government and supranational yields once again converging by late 2024.[19]

Figure 4 plots the government and supranational yields around the official SMP announcement. Prior to June 2022, both yields rose, though supranational yields increased more steeply (from 17% to 34%) than government yields (from 18% to 24%), likely reflecting the asymmetric impact of policy interventions described above. The June 2022 announcement marks a significant divergence: government yields plummeted from approximately 24% to below 10% over the following months, while supranational yields spiked above 40% before gradually declining to around 30%.

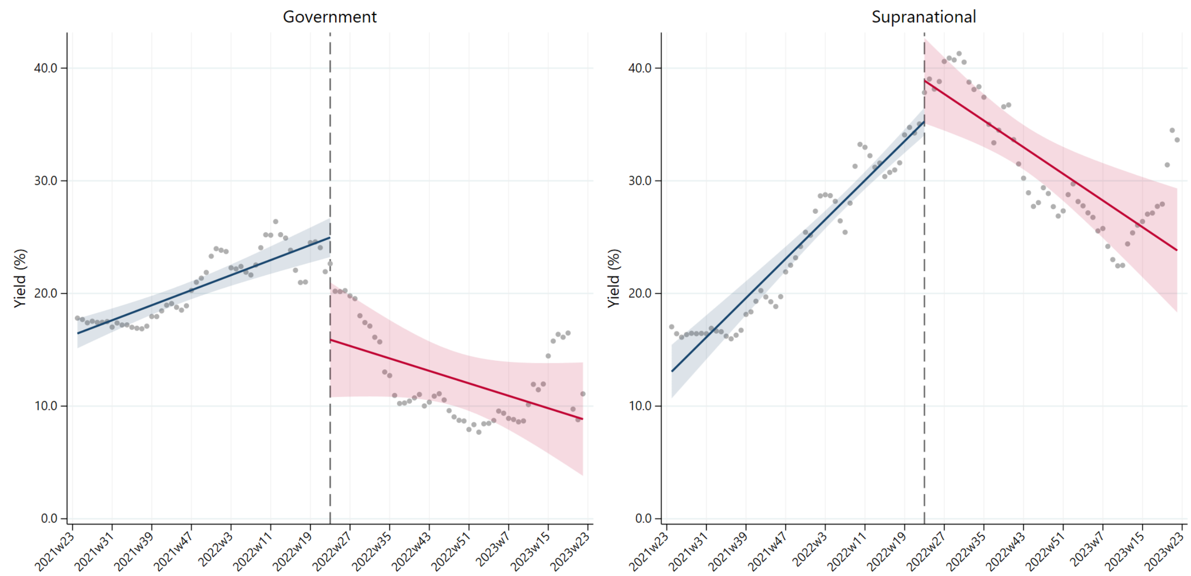
While both yields co-move after the initial shock, they remain separated by approximately 20 percentage points through late 2023. This persistent gap coincides with the period when the SMP was in effect, suggesting the policy created additional captive domestic demand that helped decouple government yields from macroeconomic fundamentals. While concurrent monetary easing and deteriorating supranational liquidity prevent attributing the entire divergence to the SMP, these patterns are consistent with financial repression suppressing government borrowing costs below the unrepressed supranational benchmarks.

[17]These foreign currency-linked accounts were guaranteed by the state and promised to pay the losses incurred on interest income due to exchange rate depreciations.

[18]The SMP was informally announced at the May 26, 2022 Monetary Policy Committee meeting, with detailed requirements published in the official gazette on June 10, 2022. The policy took effect in July 2022.

[19]Beginning June 22, 2023, authorities initiated a gradual simplification of the SMP, reducing maintenance ratios over several months before fully terminating the policy on May 9, 2024 (Celebi et al., 2025).

Figure 4: Five-year Turkish lira zero-coupon yields around the SMP announcement



Notes: This figure plots the weekly averages of daily five-year Turkish lira-denominated government (left) and supranational (right) zero-coupon yields, expressed in percent. The vertical dashed line marks the week containing 10 June 2022, when the detailed Securities Maintenance Practice requirements were published. The navy and red lines are separate pre- and post-event OLS linear fits, respectively; the event week enters both fits. Shaded regions are pointwise 95% confidence intervals for the fitted mean, calculated using Newey–West heteroskedasticity-and-autocorrelation-consistent standard errors with a Bartlett kernel, eight weekly lags, and the finite-sample adjustment. The sample covers the 51 weeks before and after the event week. Sources: Bloomberg, Refinitiv, and author's calculations.

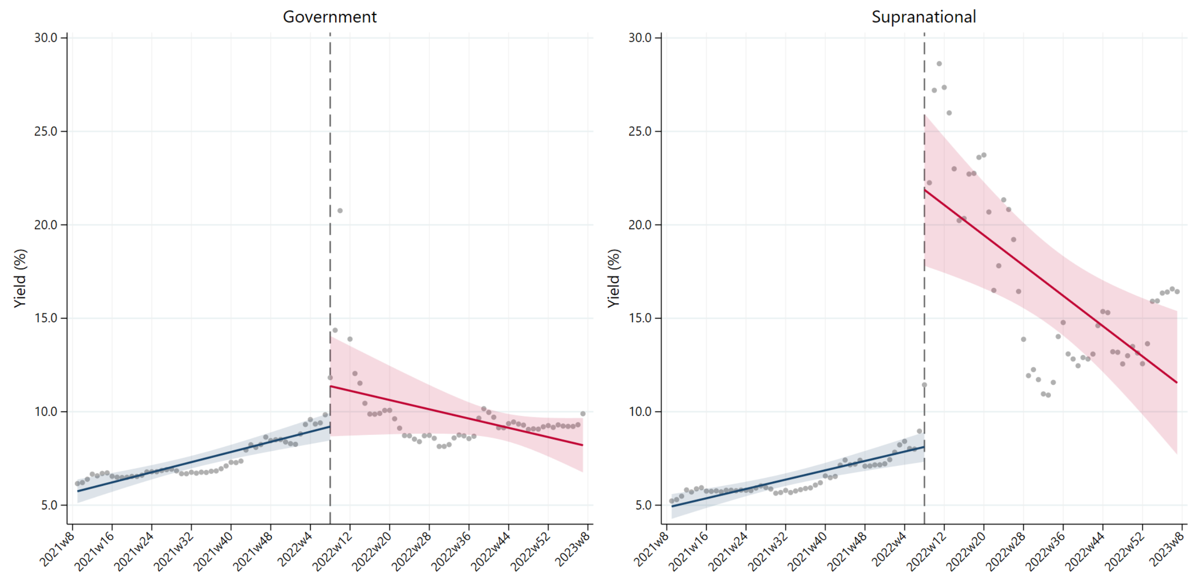
5.5.2 Russia

Figure 5 plots the Russian rouble-denominated five-year government and supranational yields around the beginning of the Russia-Ukraine war in February 2022. Prior to February 2022, both yields moved in tandem with government bonds trading slightly above supranational bonds (around 8-9% versus 7-8%, respectively). Following the war, government yields spiked to 20% before rapidly stabilising around 10%, while supranational yields surged to 30% and remained highly volatile around 15%, reversing the DBP from positive to deeply negative.

The initial government yield spike coincided with foreign investors exiting Russian LC government bonds (OFZs) – the nominal value of foreign holdings fell by 53% between January 2022 and June 2023. However, prices quickly stabilised as domestic financial institutions absorbed the excess supply in the secondary market. During this period, based on external debt statistics published by the Central Bank of Russia, domestic ownership share of OFZs reached 93%. With international sanctions blocking access to global financial markets and capital controls imposed by the government restricting currency conversion, Russian financial institutions were pushed towards rouble-denominated assets, government bonds being the primary option.[20] Meanwhile, foreign investors largely withdrew from unsanctioned rouble-denominated supranational bonds amid broader sanctions on Russian assets. Russian investors could not trade these bonds either: sanctions severed their access to international clearing networks through which many of these

[20] For information about international sanctions on Russian government debt, see Cleary Gottlieb Steen & Hamilton LLP (2023). For Russian capital controls, see Demertzis et al. (2022).

Figure 5: Five-year Russian rouble zero-coupon yields around the Russia-Ukraine war



Notes: This figure plots the weekly averages of daily five-year Russian rouble-denominated government (left) and supranational (right) zero-coupon yields, expressed in percent. The vertical dashed line marks the week containing 22 February 2022; Russia launched its full-scale invasion of Ukraine later that week. The navy and red lines are separate pre- and post-event OLS linear fits, respectively; the event week enters both fits. Shaded regions are pointwise 95% confidence intervals for the fitted mean, calculated using Newey–West heteroskedasticity-and-autocorrelation-consistent standard errors with a Bartlett kernel, eight weekly lags, and the finite-sample adjustment. The sample covers the 51 weeks before and after the event week. Sources: Bloomberg, Refinitiv, and author's calculations.

bonds settled.[21]

The Russian case shows how an exogenous geopolitical shock can create captive demand for LC government debt. While supranational bonds languished without buyers, OFZs benefited from domestic institutions having fewer alternative investment options. The resulting negative DBP quantifies how this shock transformed domestic financial institutions into forced buyers of government debt.[22] Together, the Turkey and Russia examples illustrate how powerful financial repression can be in decoupling government borrowing costs from creditworthiness.

6 Concluding remarks

This paper proposes the Domestic Bond Premium, the yield spread between EM LC government bonds and same-currency AAA-rated supranational bonds, as a high-frequency measure to track financial repression. To compute it, I construct supranational zero-coupon yield curves for 11 EM currencies from a security-level panel of more than 4,000 bonds. Because the two legs of the spread share identical currency exposure, the DBP nets out the risk-free rate and isolates sovereign default risk, relative liquidity, and financial repression components.

Empirically, the DBP is persistently low and frequently negative, remains stable during crises when EM LC yield spreads over US Treasuries widen sharply, and is driven primarily by

[21]See Kaniecki et al. (2024) on sanctions against Russian securities infrastructure including the National Settlement Depository.

[22]This pattern is consistent with Itskhoki and Mukhin (2025), who discuss that greater international financial isolation makes financial repression more potent in extracting fiscal surplus from the private sector.

local rather than global factors. Financial repression proxies are statistically and economically significant in explaining its movements, and daily local projections around discrete macroprudential policy events show the DBP moving in the directions the model predicts. The case studies of Turkey and Russia show repression holding government yields far below default-free benchmarks even as creditworthiness deteriorated.

These findings have four broader implications. First, financial repression can be tracked through asset prices. Because the DBP controls for currency risk by construction and observable proxies can account for default risk and liquidity differentials, the residual variation provides a continuous signal of repression intensity. This complements institutional classifications and retrospective analyses, which are typically available only at low frequency and with lags.

Second, the DBP isolates the non-currency component that headline EM LC government–US Treasury spreads bundle together with currency risk. The decomposition is useful in itself: the large swings in those headline spreads during stress are driven mainly by currency risk premia, while the non-currency component stays small. That small premium, however, should not be read as evidence of low repayment risk, because the paper shows it may be held down in part by captive domestic demand rather than by strong economic fundamentals.

Third, negative DBPs show that financial repression can push government borrowing costs below default-free benchmarks across the maturity spectrum. Unlike convenience yields on US Treasuries, which reflect safe-haven properties, these negative spreads potentially rest on captive demand that can reverse abruptly, and any resulting fiscal relief may come at the cost of crowding out private investment.

Fourth, expanded supranational issuance could erode the repression premium by providing domestic investors an alternative offshore LC asset. Governments can, however, restrict access through unfavourable regulatory treatment, so whether international institutional design can effectively constrain repression remains an open question.

Several limitations temper these conclusions and point to future work. Supranational yield curves are estimated from bonds that trade at irregular maturities with varying liquidity, which may introduce measurement error in DBP estimation, and financial repression proxies remain imperfect. The panel regression estimates and the local projections capture correlations rather than causal effects; causal identification would require additional structure exploiting policy variation that is plausibly exogenous to the business cycle. Finally, understanding how institutional quality and external financing constraints shape governments' willingness to create captive demand could help predict when repression emerges and when it becomes unsustainable.

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Appendix

A Model derivations

This appendix derives the domestic investor's Euler equations, the log-linear yield decompositions, and the ownership-share result underlying the framework in Section 2.

A.1 Domestic investor problem and first-order conditions

The representative domestic investor chooses $\{c_t, B_{t+1}^{D,Gov}, B_{t+1}^{D,Sup}\}_{t\geq 0}$ to maximise

$$U_0 \;=\; \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big[u(c_t) + v\big(B_{t+1}^{D,Gov} + B_{t+1}^{D,Sup}\big)\Big], \tag{33}$$

subject to the period budget constraint

$$c_t + (P_t^{Gov} + \varphi_t^{Gov})B_{t+1}^{D,Gov} + (P_t^{Sup} + \varphi_t^{Sup})B_{t+1}^{D,Sup} = \omega + (1-\delta_t)B_t^{D,Gov} + B_t^{D,Sup} \;\equiv\; W_t, \tag{34}$$

and the minimum-holdings (financial repression) constraint

$$P_t^{Gov} B_{t+1}^{D,Gov} \;\geq\; \kappa_t W_t, \qquad \kappa_t \in [0,1], \tag{35}$$

where $\delta_{t+1} \in [0,1]$ denotes the (random) haircut on government debt realised at $t+1$.
Let $\mu_t > 0$ be the multiplier on (34) and $\lambda_t \geq 0$ the multiplier on (35). The Lagrangian is

$$\begin{aligned}
\mathcal{L}_0 = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Big[& u(c_t) + v\big(B_{t+1}^{D,Gov} + B_{t+1}^{D,Sup}\big) \\
& + \mu_t\big(W_t - c_t - (P_t^{Gov} + \varphi_t^{Gov})B_{t+1}^{D,Gov} - (P_t^{Sup} + \varphi_t^{Sup})B_{t+1}^{D,Sup}\big) \\
& + \lambda_t\big(P_t^{Gov} B_{t+1}^{D,Gov} - \kappa_t W_t\big)\Big],
\end{aligned} \tag{36}$$

with complementary slackness

$$\lambda_t \geq 0, \quad P_t^{Gov} B_{t+1}^{D,Gov} - \kappa_t W_t \geq 0, \quad \lambda_t\big(P_t^{Gov} B_{t+1}^{D,Gov} - \kappa_t W_t\big) = 0. \tag{37}$$

FOC for consumption.

$$u'(c_t) - \mu_t = 0 \quad \Rightarrow \quad \mu_t = u'(c_t). \tag{38}$$

FOC for supranational bonds. The choice $B_{t+1}^{D,Sup}$ affects: (i) utility at t, (ii) the period-t budget, (iii) next-period wealth W_{t+1} (hence the $t+1$ repression constraint), and (iv) the $t+1$ budget. The FOC is

$$v'(T_{t+1}) - \mu_t(P_t^{Sup} + \varphi_t^{Sup}) + \beta\, \mathbb{E}_t[\mu_{t+1} - \lambda_{t+1}\kappa_{t+1}] = 0. \tag{39}$$

FOC for government bonds. The choice $B_{t+1}^{D,Gov}$ affects: (i) utility at t, (ii) the period-t budget, (iii) the period-t repression constraint directly, (iv) next-period wealth via the payoff

$(1-\delta_{t+1})B_{t+1}^{D,Gov}$ (hence the $t+1$ repression constraint), and (v) the $t+1$ budget. The FOC is

$$v'(T_{t+1}) - \mu_t(P_t^{Gov} + \varphi_t^{Gov}) + \lambda_t P_t^{Gov} + \beta\,\mathbb{E}_t\big[\big(\mu_{t+1} - \lambda_{t+1}\kappa_{t+1}\big)(1-\delta_{t+1})\big] = 0. \tag{40}$$

Because convenience enters through the sum $T_{t+1} = B_{t+1}^{D,Gov} + B_{t+1}^{D,Sup}$, the same marginal service value $v'(T_{t+1})$ appears in (39) and (40).

A.2 Constrained pricing kernel and Euler (pricing) equations

Define the shadow marginal value of wealth

$$\tilde{\mu}_{t+1} \;\equiv\; \mu_{t+1} - \lambda_{t+1}\kappa_{t+1}, \tag{41}$$

and the constrained pricing kernel

$$\tilde{M}_{t+1} \;\equiv\; \beta\frac{\tilde{\mu}_{t+1}}{\mu_t} = \beta\frac{u'(c_{t+1}) - \lambda_{t+1}\kappa_{t+1}}{u'(c_t)}. \tag{42}$$

Also define the (scaled) shadow value of the repression constraint

$$\rho_t \;\equiv\; \frac{\lambda_t}{u'(c_t)} \;\geq 0, \tag{43}$$

and the marginal convenience of LC bonds in marginal utility units

$$\chi_t \;\equiv\; \frac{v'(T_{t+1})}{u'(c_t)}. \tag{44}$$

Dividing (39)–(40) by $\mu_t = u'(c_t)$ yields the Euler (pricing) equations:

$$P_t^{Sup} + \varphi_t^{Sup} = \mathbb{E}_t\Big[\tilde{M}_{t+1}\Big] + \chi_t, \tag{45}$$

$$P_t^{Gov} + \varphi_t^{Gov} = \mathbb{E}_t\Big[\tilde{M}_{t+1}(1-\delta_{t+1})\Big] + \chi_t + \rho_t P_t^{Gov}. \tag{46}$$

A.3 Bond prices

From (45),

$$P_t^{Sup} = \underbrace{\mathbb{E}_t[\tilde{M}_{t+1}]}_{\tilde{q}_t} + \chi_t - \varphi_t^{Sup}. \tag{47}$$

From (46), collecting terms in P_t^{Gov} gives

$$P_t^{Gov} = \frac{\mathbb{E}_t[\tilde{M}_{t+1}(1-\delta_{t+1})] + \chi_t - \varphi_t^{Gov}}{1-\rho_t}. \tag{48}$$

A.4 Log-linear yield decompositions

Define continuously-compounded one-period yields

$$y_t^b \equiv -\log P_t^b, \qquad b \in \{Gov, Sup\}. \tag{49}$$

Let

$$\tilde{q}_t \equiv \mathbb{E}_t[\tilde{M}_{t+1}], \qquad \tilde{r}_t \equiv -\log(\tilde{q}_t), \tag{50}$$

denote the shadow risk-free asset price and rate for the constrained domestic investor.

Supranational yield. Using (47), $P_t^{Sup} = \tilde{q}_t + (\chi_t - \varphi_t^{Sup})$. A first-order expansion of $-\log(\tilde{q}_t + x)$ around $x = 0$ gives

$$y_t^{Sup} \approx \tilde{r}_t + \underbrace{\frac{\varphi_t^{Sup}}{\tilde{q}_t}}_{\text{liquidity}} + \underbrace{\left(-\frac{\chi_t}{\tilde{q}_t}\right)}_{\text{private convenience}} . \tag{51}$$

Government yield. Let

$$D_t \equiv \mathbb{E}_t[\tilde{M}_{t+1}\delta_{t+1}], \quad \Rightarrow \quad \mathbb{E}_t[\tilde{M}_{t+1}(1-\delta_{t+1})] = \tilde{q}_t - D_t. \tag{52}$$

Then (48) implies

$$P_t^{Gov} = \frac{\tilde{q}_t + (-D_t + \chi_t - \varphi_t^{Gov})}{1-\rho_t}. \tag{53}$$

Taking logs,

$$\log P_t^{Gov} = \log\big(\tilde{q}_t + x_t\big) - \log(1-\rho_t), \qquad x_t \equiv -D_t + \chi_t - \varphi_t^{Gov}. \tag{54}$$

Using first-order approximations $\log(\tilde{q}_t + x_t) \approx \log \tilde{q}_t + x_t/\tilde{q}_t$ and $-\log(1-\rho_t) \approx \rho_t$ yields

$$y_t^{Gov} \approx \tilde{r}_t + \underbrace{\frac{D_t}{\tilde{q}_t}}_{\text{default}} + \underbrace{\frac{\varphi_t^{Gov}}{\tilde{q}_t}}_{\text{liquidity}} + \underbrace{\left(-\frac{\chi_t}{\tilde{q}_t} - \rho_t\right)}_{\text{private convenience \& repression}} . \tag{55}$$

It is convenient to define the (model-implied) default-loss premium

$$I_t^{Gov} \equiv \frac{\mathbb{E}_t[\tilde{M}_{t+1}\delta_{t+1}]}{\mathbb{E}_t[\tilde{M}_{t+1}]} = \frac{D_t}{\tilde{q}_t}, \tag{56}$$

so that (55) can be written as $y_t^{Gov} \approx \tilde{r}_t + I_t^{Gov} + \varphi_t^{Gov}/\tilde{q}_t - \chi_t/\tilde{q}_t - \rho_t$.

A.5 Domestic bond premium

The Domestic Bond Premium is the same-currency spread

$$DBP_t \equiv y_t^{Gov} - y_t^{Sup}. \tag{57}$$

Using (51) and (55), the shadow risk-free rate $\tilde{r}_t$ and the convenience term $\chi_t/\tilde{q}_t$ cancel, giving

$$DBP_t \approx \underbrace{I_t^{Gov}}_{\text{default}} + \underbrace{\frac{\varphi_t^{Gov} - \varphi_t^{Sup}}{\tilde{q}_t}}_{\text{relative liquidity}} - \underbrace{\rho_t}_{\text{repression wedge}} , \tag{58}$$

which is (30) in the text.

A.6 The ownership share

In the binding regime, substituting $B_{t+1}^{D,Gov} = s_t^D \bar{B}_t^{Gov}$ and the foreign-side price (20) into the constraint $P_t^{Gov} B_{t+1}^{D,Gov} = \kappa_t W_t$ gives (25), i.e. $H(s_t^D) = \kappa_t W_t$ with

$$H(s) \equiv s\,\bar{B}_t^{Gov}\Big[V_t^{F,Gov} - \psi_t^{Gov}(1-s)\bar{B}_t^{Gov}\Big], \qquad H'(s) = \bar{B}_t^{Gov}\Big[P_t^{Gov} + s\,\psi_t^{Gov}\bar{B}_t^{Gov}\Big] > 0. \quad (59)$$

By the implicit function theorem,

$$\frac{\partial s_t^D}{\partial \kappa_t} = \frac{W_t}{H'(s_t^D)} > 0, \qquad \frac{\partial s_t^D}{\partial V_t^{F,Gov}} = -\frac{s_t^D\,\bar{B}_t^{Gov}}{H'(s_t^D)} < 0, \qquad \frac{\partial s_t^D}{\partial \psi_t^{Gov}} = \frac{s_t^D(1-s_t^D)(\bar{B}_t^{Gov})^2}{H'(s_t^D)} > 0. \quad (60)$$

Higher repression intensity raises the domestic share, stronger foreign demand lowers it, and weaker foreign capacity raises it.

B DBP vs LCCS: conceptual and empirical differences

Is the DBP the only measure that captures financial repression? This appendix compares it to the local currency credit spread (LCCS) of Du and Schreger (2016) and argues the latter might confound domestic policy-induced distortions in government bond markets, as it also incorporates CIP deviations and the US Treasury premium.

B.1 Currency-hedged US Treasuries as a benchmark

Fix currency j and tenor τ. Let $f_{j,t,\tau}$ denote the forward premium for converting dollars into currency j at maturity τ (i.e., the cost of hedging the dollar payoff into currency j). The LCCS is defined as

$$LCCS_{j,\$,t,\tau} \;\equiv\; y_{j,t,\tau}^{Gov} - \big(y_{\$,t,\tau}^{Gov} + f_{j,t,\tau}\big), \quad (61)$$

where $y_{\$,t,\tau}^{Gov}$ is the U.S. Treasury zero-coupon yield. Then, $y_{\$,t,\tau}^{Gov} + f_{j,t,\tau}$ is the yield on a currency-hedged US Treasury, expressed in currency j units. Using the yield decomposition (11), the LCCS can be expressed as

$$LCCS_{j,\$,t,\tau} \;\equiv\; \big(I_{j,t,\tau}^{Gov} - I_{\$,t,\tau}^{Gov}\big) + \big(L_{j,t,\tau}^{Gov} - L_{\$,t,\tau}^{Gov}\big) + \big(R_{j,t,\tau}^{Gov} - R_{\$,t,\tau}^{Gov}\big) - \; CIP_{j,t,\tau}, \quad (62)$$

where the CIP basis is

$$CIP_{j,t,\tau} \;\equiv\; \big(\tilde{r}_{\$,t,\tau} + f_{j,t,\tau}\big) - \tilde{r}_{j,t,\tau}. \quad (63)$$

When $CIP_{j,t,\tau} \approx 0$, LCCS indeed isolates the non-currency risk premia on EM LC government bonds.[23] However, Dao and Gourinchas (2026) document persistent CIP deviations in EMs since the early 2000s. Moreover, even if currency hedging markets are frictionless, the potentially large convenience yield on US Treasuries (e.g., Krishnamurthy and Vissing-Jorgensen,

[23] See Appendix B.2 for a simple extension of the model that demonstrates when the CIP condition can fail.

2012; Du et al., 2018) may confound EM issuer-specific non-currency risk premia. For these reasons, I focus on DBP as the primary high-frequency measure to track financial repression.

B.2 Dollar funding frictions

To see why the CIP condition might fail, consider a competitive financial intermediary that clears the FX forward market at tenor τ for the (j, USD) currency pair via swap contracts. Let $H_{j,t,\tau}$ denote the time-t net notional demand to buy USD forward against currency j (i.e., to receive USD at time $t+\tau$ by delivering currency j at time $t+\tau$). Thus, $H_{j,t,\tau} > 0$ means investors want to buy USD forwards (so the intermediary must sell USD forwards) and $H_{j,t,\tau} < 0$ the opposite.

Although at time t the forward position is off-balance-sheet for the financial intermediary (it realises only at time $t + \tau$), taking net notional exposure $|H_{j,t,\tau}|$ requires posting an initial margin: a fraction $\delta_{t,j,\tau} \in (0, 1)$ of the notional must be earmarked on the balance sheet. Let ϑ_t denote the shadow value of balance sheet capacity (the Lagrange multiplier on the intermediary's balance sheet constraint). Then supplying one additional unit of forward notional uses $\delta_{t,j,\tau}$ units of balance sheet capacity and thus has marginal cost

$$MC^{FX}_{t,j,\tau} = \vartheta_t \, \delta_{t,j,\tau}. \tag{64}$$

In this stylised illustration, only the initial margin on FX forward positions consumes balance sheet capacity; the spot borrowing and lending legs of the hedging strategy described below are assumed to be unconstrained.

Note that the intermediary can eliminate exchange-rate risk on its net forward position by applying a simple trading strategy at time t: (i) sell one unit of USD notional forward, (ii) borrow in currency j at rate $\tilde{r}_{j,t,\tau}$, (iii) lend in USD at rate $\tilde{r}_{\$,t,\tau}$. By doing so, it fixes the period $t + \tau$ FX-rate and receives a currency-hedged payoff:

$$MR^{FX}_{j,t,\tau} = \tilde{r}_{\$,t,\tau} + f_{j,t,\tau} - \tilde{r}_{j,t,\tau},$$

which is identical to the CIP deviation formula. Since the intermediary prices swap contracts fairly, we have the following identity

$$MC^{FX}_{t,j,\tau} = MR^{FX}_{j,t,\tau} \iff \vartheta_t \delta_{t,j,\tau} = CIP_{j,t,\tau}. \tag{65}$$

Therefore, if the intermediary's balance sheet constraint becomes binding ($\vartheta_t > 0$) and initial margin requirements rise (higher $\delta_{t,j,\tau}$) or net dollar-hedging demand rises (higher $|H_{j,t,\tau}|$), the CIP basis can be non-zero.

This simple framework demonstrates why the LCCS is a confounded measure of EM-specific non-currency risk premia. First, because it subtracts a currency-hedged U.S. Treasury benchmark, the LCCS loads on $CIP_{j,t,\tau}$, which is driven by global financial intermediary constraints and dollar-hedging demand. Second, even if the FX swap market is frictionless and $CIP_{t,j,\tau} \approx 0$, the LCCS still inherits the time variation in the U.S. Treasury pricing component $R^{Gov}_{\$,t,\tau}$. This component may reflect flight-to-safety dynamics that intensify during global risk-off periods.

B.3 Empirical comparison

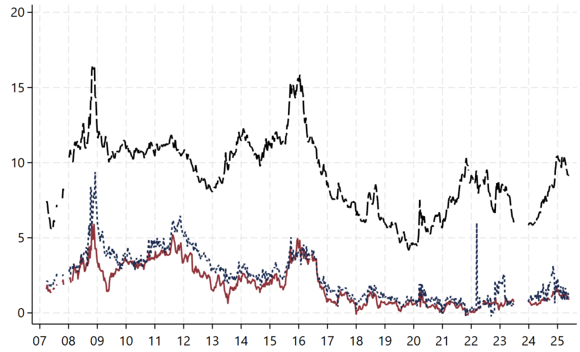
Figure 6 compares the five-year DBP with the LCCS defined in Appendix B.1. During the sample period when they are both available, the DBP and LCCS track each other closely (correlation ≈ 0.93) and both lie well below the EM LC government–US Treasury spread, except in China after 2022, where policy rates were cut while other major economies tightened aggressively. This suggests that DBP and LCCS capture similar risk components.

However, the DBP tends to lie below the LCCS in most countries and time periods, and the two measures occasionally diverge during stress episodes (most notably in Brazil, South Africa, and Mexico during the GFC).

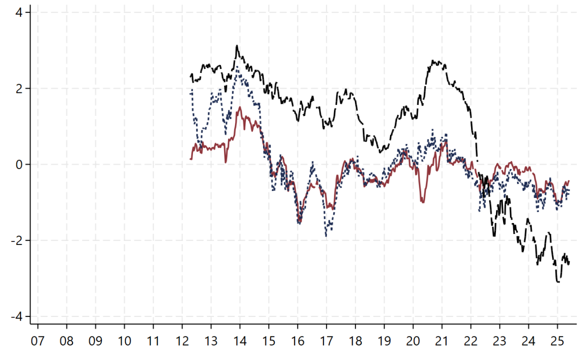
As explained in Section 2 and Appendix B.2, LCCS embeds CIP deviations and the US Treasury convenience yield, both of which are persistent and can account for the LCCS lying modestly above the DBP even in normal times. These components widen during stress episodes: CIP deviations increase as financial intermediary balance sheets become more constrained (Du and Schreger, 2022a), and Treasury yields decline relative to other developed market sovereigns as investors seek safe havens (Du et al., 2018), mechanically inflating the LCCS even if EM yields remain unchanged. Additionally, forward contracts for EM currencies vary in structure across countries and time (e.g., deliverable versus non-deliverable) and can become highly illiquid during financial stress, contributing further noise to the LCCS.

Naturally, the DBP could also be affected by the liquidity differentials between government and supranational bonds, but this is relatively straightforward to control in regressions using proxies for secondary-market liquidity. Therefore, the nature of measurement challenges in both spreads differs significantly: the LCCS may be influenced by global phenomena such as CIP deviations and US Treasury premium that affect all EMs simultaneously, while frictions embedded in the DBP tend to be idiosyncratic and amenable to direct measurement through secondary-market liquidity proxies.

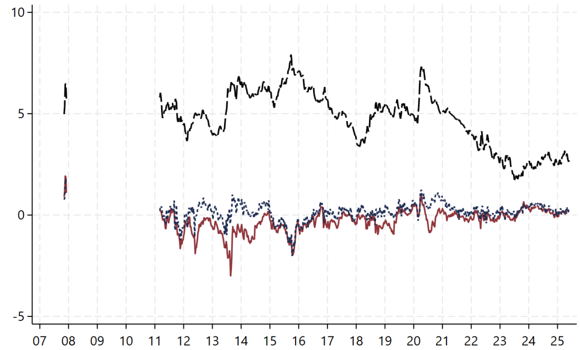
Figure 6: Five-year yield spreads (DBP, LCCS, and EM LC government–US Treasury)



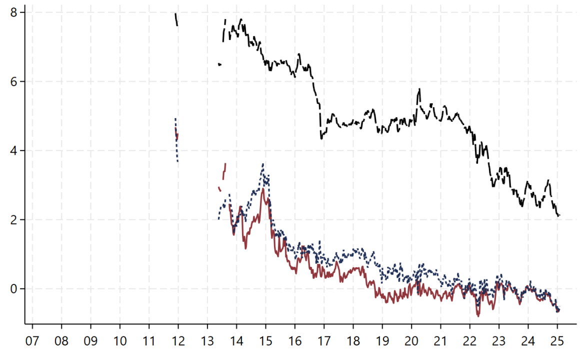
(a) Brazilian real



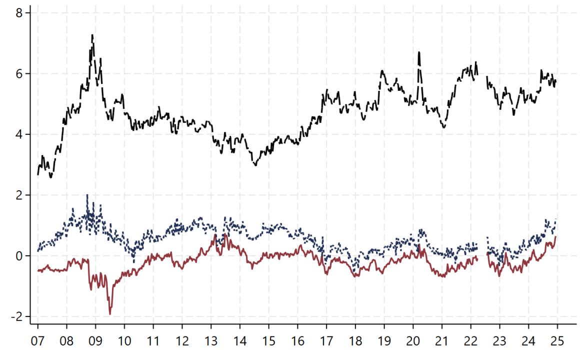
(b) Chinese renminbi



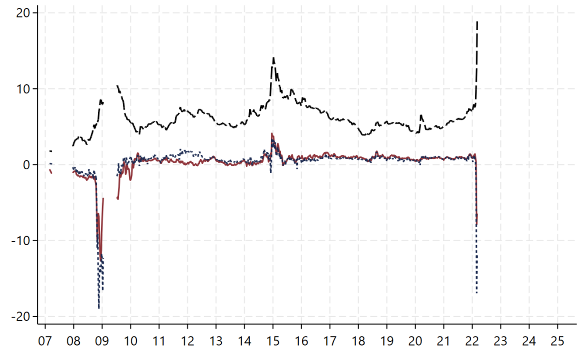
(c) Indonesian rupiah



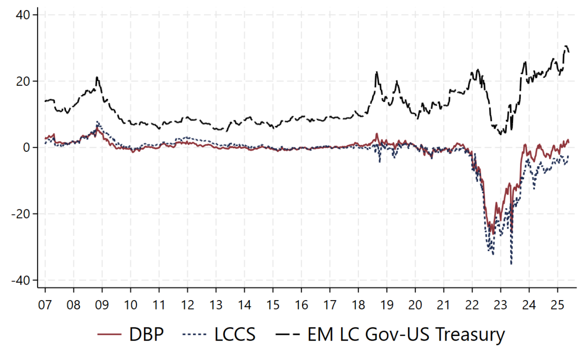
(d) Indian rupee



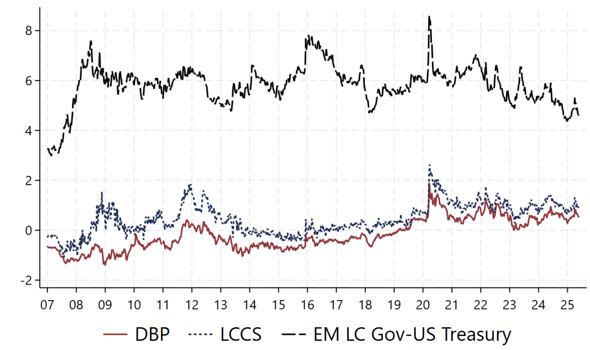
(e) Mexican peso



(f) Russian rouble



(g) Turkish lira



(h) South African rand

Notes: This figure plots the DBP (solid red line), LCCS (dotted blue line), and EM local-currency government–US Treasury spread (dashed black line) at the five-year tenor, expressed in percentage points, for eight EM currencies. Each panel retains dates for which all three measures are available. Sources: Bloomberg, Refinitiv, Du and Schreger (2016) (updated through 2025), and author's calculations.

C Data sources

Table 7: Variables and data sources

Variable	Source
Supranational bond prices and characteristics	Bloomberg
Government zero-coupon yields	Bloomberg; Refinitiv
Sovereign CDS; VIX; CPI; Equity indices	Refinitiv (Datastream)
Spot exchange rates	Bank for International Settlements
Policy interest rates	Bank for International Settlements
EM currency forward premiums	Du and Schreger (2016) (updated, 2025)
Sovereign debt holdings; Debt-to-GDP	Arslanalp and Tsuda (2014) (updated, 2025)

D Term-structure model for supranational yield curves

Bond-level price panel and preprocessing. I estimate the supranational benchmark curve separately for each currency using the underlying panel of individual bond prices. The estimation sample is restricted to fixed-rate coupon and zero-coupon bonds; inflation-linked, floating-rate, callable, and putable securities are excluded. For each bond-date observation, I reconstruct the full schedule of remaining cash flows from the issue date, maturity date, coupon rate, payment frequency, day-count convention, and end-of-month rule, and convert the quoted clean-mid-price into a dirty price by adding accrued interest.

Before estimation, I apply several screens designed to remove observations that are unlikely to be informative about the cross-section of discount rates. First, I exclude January 1, December 25, and December 26, when trading activity is often exceptionally thin. Second, I drop observations with remaining maturity below 30 days, as well as rows with missing or non-positive clean prices. Third, when Bloomberg reports a yield-to-maturity for the bond, I trim observations outside the 0.5th and 99.5th percentiles of the within-currency yield distribution. Fourth, I remove "partial-market" dates on which the number of surviving bond quotes collapses relative to nearby dates within the same currency. Finally, I apply a bond-level erratic-quote screen that flags prices whose time-series behaviour is inconsistent with both nearby observations for the same bond and same-date peers in yield space; flagged prices are repaired by log-price interpolation when feasible and otherwise dropped. The resulting cleaned panel is then used to estimate a daily zero-coupon curve for each currency.

Price-based Nelson-Siegel representation. Let $\boldsymbol{\beta}_{j,t} \equiv [\beta^L_{j,t}\ \beta^S_{j,t}\ \beta^C_{j,t}]'$ denote the level, slope, and curvature factors for currency j on date t. I model the continuously-compounded supranational zero rate at maturity τ using the standard Nelson-Siegel basis:

$$r^{Sup}_{j,t}(\tau) = \beta^L_{j,t} + \beta^S_{j,t}\left(\frac{1-e^{-\lambda\tau}}{\lambda\tau}\right) + \beta^C_{j,t}\left[\left(\frac{1-e^{-\lambda\tau}}{\lambda\tau}\right) - e^{-\lambda\tau}\right]. \tag{66}$$

I fix the decay parameter at $\lambda = 0.7308$, which corresponds to the Diebold and Li (2006) benchmark and places the curvature hump at roughly 30 months.[24] Fixing λ is useful in this application because the daily supranational bond cross-section is often too thin to identify factor loadings and latent factors jointly with precision.

Because the underlying data are bond prices rather than zero-coupon yields at fixed maturities, the measurement equation is written directly in price space. Let bond i in currency j have remaining cash flows $\{CF_{j,i,t,k}\}_{k=1}^{K_{j,i,t}}$ at times $\{s_{j,i,t,k}\}_{k=1}^{K_{j,i,t}}$, measured in years from the settlement date. The model-implied dirty price is

$$P_{j,i,t}^{dirty} = h_{j,i}(\boldsymbol{\beta}_{j,t}) + \varepsilon_{j,i,t} \equiv \sum_{k=1}^{K_{j,i,t}} CF_{j,i,t,k} \exp\big(-s_{j,i,t,k}\, r_{j,t}^{Sup}(s_{j,i,t,k})\big) + \varepsilon_{j,i,t}. \tag{67}$$

This formulation uses the full cash-flow structure of each coupon bond and avoids the need to first compress the raw data into synthetic yields at representative maturities.

State dynamics and observation noise. The latent factors evolve according to a mean-reverting diagonal AR(1) process,

$$\boldsymbol{\beta}_{j,t} = \boldsymbol{\mu}_j + A_j\big(\boldsymbol{\beta}_{j,t-1} - \boldsymbol{\mu}_j\big) + \boldsymbol{\eta}_{j,t}, \qquad A_j = \mathrm{diag}(\phi_j^L, \phi_j^S, \phi_j^C), \qquad \boldsymbol{\eta}_{j,t} \sim \mathcal{N}(\mathbf{0}, Q_j), \tag{68}$$

where Q_j is a full 3×3 covariance matrix. Relative to an unrestricted VAR, the diagonal transition matrix is a stabilising restriction that is helpful when the daily cross-section of supranational bonds is sparse, while the full innovation covariance still allows the three factors to co-move. The measurement errors are assumed conditionally Gaussian and diagonal in price space:

$$\varepsilon_{j,t} \sim \mathcal{N}(\mathbf{0}, R_{j,t}), \qquad R_{j,t} = \mathrm{diag}\big(\sigma_{j,1,t}^2, \ldots, \sigma_{j,N_{j,t},t}^2\big), \tag{69}$$

with

$$\sigma_{j,i,t} = \max\left\{\sigma_{p,j},\ \frac{1}{2} BAS_{j,i,t}\right\}, \tag{70}$$

where $\sigma_{p,j}$ is a currency-specific baseline price-noise parameter and $BAS_{j,i,t}$ is the observed bid–ask spread when available. Intuitively, this treats illiquid bonds as noisier measurements of the underlying zero curve. Because the pricing equation in (67) is nonlinear in the factors, the filter uses the Jacobian of the bond-price function with respect to $\boldsymbol{\beta}_{j,t}$ and therefore takes the form of an extended Kalman filter.

Initialization, filtering, and smoothing. I initialise the dynamic model with a sequence of static cross-sectional Nelson-Siegel fits in price space. Specifically, on each usable date, I solve

$$\widehat{\boldsymbol{\beta}}_{j,t}^{seed} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{N_{j,t}} \left[P_{j,i,t}^{dirty} - h_{j,i}(\boldsymbol{\beta})\right]^2. \tag{71}$$

[24] The median remaining maturity in the sample is around 2-2.5 years for most currencies, which justifies the choice of λ.

These seed estimates are then placed on a business-day state grid running from the first to the last observed quote date, excluding January 1, December 25, and December 26. Missing weekdays are filled by time interpolation for initialization. I then estimate the long-run mean $\boldsymbol{\mu}_j$ and the persistence parameters $(\phi_j^L, \phi_j^S, \phi_j^C)$ from factor-by-factor AR(1) regressions on the seed series, set Q_j equal to the covariance matrix of the resulting transition residuals, and estimate the baseline price-noise parameter $\sigma_{p,j}$ from the static price residuals.

Given these starting values, I run an extended Kalman filter on the full business-day state grid and then apply a Rauch-Tung-Striebel smoother to obtain the final latent factor estimates.

Reported zero-coupon yields. After smoothing, I evaluate the Nelson-Siegel curve on a fixed maturity grid. The empirical analysis in the paper uses the 1, 3, 5, and 10-year points. Let $\widehat{r}_{j,t,\tau}^{Sup}$ denote the smoothed continuously-compounded zero rate implied by (66). The reported annualised zero-coupon yield is

$$\widehat{y}_{j,t,\tau}^{Sup} = \exp\bigl(\widehat{r}_{j,t,\tau}^{Sup}\bigr) - 1. \tag{72}$$

These supranational zero-coupon yields are then matched to the corresponding government zero-coupon yields to construct

$$DBP_{j,t,\tau} = y_{j,t,\tau}^{Gov} - \widehat{y}_{j,t,\tau}^{Sup}.$$

E Variance decomposition methodology

The decomposition presented in Section 5.1 proceeds in two steps. First, I quantify how much of the spread variance reflects persistent cross-country differences attributable to slow-moving variables (e.g., institutional quality). These variables determine the average DBP in each currency. Consider a fixed tenor length τ and regress $DBP_{j,t,\tau}$ on currency fixed-effects (α_j) only:

$$DBP_{j,t,\tau} \;=\; \alpha_j \;+\; \varepsilon_{j,t,\tau}. \tag{73}$$

In this panel regression, the overall R^2 captures the between-variance R_B^2 – the share of variation explained by static currency/country characteristics. The complement $R_W^2 \equiv 1 - R_B^2$ represents the within-variance: the share of DBP variation explained by transitory variables such as fiscal and monetary shocks, regulatory interventions, or evolving economic conditions.

Next, I distinguish between global factors that affect all currencies simultaneously and country-specific factors. To isolate time-varying components, I first demean the data at the currency-tenor level. Let $T_{j,\tau}$ denote the number of time-series observations in a currency-tenor pair. I compute currency-tenor means $\overline{DBP}_{j,\tau} = \frac{1}{T_{j,\tau}} \sum_t DBP_{j,t,\tau}$ and define the centred values as $\widetilde{DBP}_{j,t,\tau} = DBP_{j,t,\tau} - \overline{DBP}_{j,\tau}$. Regressing these centred values on time fixed effects (ω_t) captures (average) shocks common to all currencies:

$$\widetilde{DBP}_{j,t,\tau} \;=\; \omega_t \;+\; \xi_{j,t,\tau}. \tag{74}$$

Let R_{WG}^2 be the overall R^2 from (74). Since the centred values in (74) contain only within-variation, the within-variance share can be decomposed as:

$$\text{Global factors: } R^2_G = R^2_W \times R^2_{WG} \quad \text{and} \quad \text{Local factors: } R^2_L = R^2_W \times \left(1 - R^2_{WG}\right),$$

so that $R^2_B + R^2_G + R^2_L = 100\%$ of the variance by construction.

F iMaPP policy events: construction and pre-event diagnostic

Source data and sample. The iMaPP database (Alam et al., 2019) records macroprudential policy actions across 134 countries from 1990 onward, comprising both monthly tightening–loosening dummies and narrative descriptions of each action across 17 instrument categories. For the 11 sample currencies and the period from 2005 onward[25], I retain narrative cells for the parent instrument categories: reserve requirements (RR), liquidity-based measures (Liquidity), capital-based measures (Capital), restrictions on foreign-currency lending (LFC), limits on FX positions (LFX), restrictions on consumer or corporate loans (LoanR), leverage requirements (LVR), other macroprudential instruments (OT), and bank levies and taxes (Tax). This yields 767 narrative cells.

Classification. The structured extraction described in Section 5.4 requires the model to flag whether each focal action is relevant for government bond demand and, for relevant actions, to assign a direction label and a transmission channel. The direction label takes three values: U for actions that increase equilibrium demand for government bonds, D for actions that decrease demand, and A for relevant actions whose direction cannot be inferred from the text or that involve no substantive change (for example, technical re-issuance of an existing rule). A-labelled events are emitted for completeness but do not enter the local projection in equation (32). Actions classified as not relevant for government bond demand, typically standard prudential measures without a sovereign-specific dimension (e.g., leverage requirements or borrower-side loan restrictions), form the neutral events that enter equation (32) through the $N_{j,t}$ indicator as a placebo. The channel label takes four values: *mandate* captures direct requirements to hold government securities, such as securities-maintenance practices or statutory liquidity ratios; *indirect* captures measures that narrow non-sovereign or non-LC alternatives without an explicit government bond quota, such as reserve requirements on FC deposits; *foreign* captures measures restricting foreign-investor access to the LC sovereign bond market, such as currency convertibility limits or derivatives counterparty restrictions; and *other* is a residual category. To preserve an audit trail at the event level, the model is required to quote the phrase from the source text justifying its direction and channel labels. From the 767 narrative cells, the procedure yields 1,129 distinct focal actions, of which 602 are classified as relevant for government bond demand; the remaining 527 constitute the pool of neutral events.

Classification prompt. The system prompt given to the large language model (Anthropic Claude Sonnet 4.6) is as follows.

[25] I retain the period from 2005 onwards to account for the events that may be announced before 2007 but are implemented a few years later.

```
You are a policy analyst extracting individual macroprudential policy actions
↪  from IMF iMaPP database narrative cells.

For each focal action, report:
- whether it is RELEVANT to equilibrium demand for local-currency (LC) government
↪  bonds,
- when relevant, the DIRECTION and CHANNEL of the effect,
- a short action summary and a reasoning quote from the text.

RELEVANT (boolean):
- true  - the action concerns equilibrium demand for LC government bonds, for
↪  example by mobilising captive domestic demand, by restricting
↪  foreign-investor access to the LC sovereign-bond market, or by restricting
↪  domestic access to foreign assets or non-sovereign alternatives.
- false - the action is outside this scope. Examples: pure household
↪  borrower-side rules, capital requirement restrictions without specific
↪  mention of sovereign treatment, generic loss-provisioning or accounting
↪  rules; framework-only publications; methodology notes.
Still emit the event when false, so we can use these actions as a placebo.

DIRECTION - populate ONLY when relevant=true; null otherwise.
- U (up): the action increases demand for sovereign debt. Examples: a new
↪  requirement that banks hold LC government securities through reserve or
↪  liquidity requirements; reserve-requirement hikes on FC deposits that push
↪  banks toward LC assets; risk-weight hikes on non-sovereign LC assets that
↪  make sovereigns relatively more attractive or risk-weight reductions on
↪  sovereign bonds; capital controls or FX-position caps that trap domestic
↪  savings or push foreign holders out, etc.
- D (down): the action decreases demand. Examples: termination or relaxation of a
↪  securities-holding mandate; suspension of a minimum LCR or NSFR (reduces HQLA
↪  demand for sovereigns); removal of capital controls on portfolio outflows;
↪  lowering a risk-weight OR taxation preferential treatment for sovereign
↪  holdings.
- A (ambiguous or neutral): a focal action whose net direction on LC sovereign
↪  bond demand cannot be signed - either because the text does not reveal
↪  direction, or because the action reaffirms / maintains / re-issues an
↪  existing rule with no substantive change (e.g., "reserve requirements were
↪  restructured"; "the Board confirmed the MPL at its current level").

CHANNEL - populate ONLY when relevant=true; null otherwise.
- mandate: a direct requirement that domestic institutions hold government
↪  securities. Examples: securities-maintenance rules; minimum LCR with
↪  sovereigns counted as HQLA; sovereign treatment in capital-adequacy
↪  frameworks.
```

```
- indirect: measures that narrow domestic institutions' non-sovereign or non
↪  local-currency denominated alternatives without an explicit sovereign bond
↪  quota. Examples: reserve requirements on LC or FC deposits or loans; limits
↪  on FC lending; FX-position caps on banks; risk-weight hikes on non-sovereign
↪  LC assets. etc.
- foreign: measures restricting foreign-investor access to the LC sovereign-bond
↪  market. Examples: capital controls on portfolio inflows or outflows;
↪  restrictions on non-residents' derivative transactions with domestic banks;
↪  currency-convertibility limits.
- other: clearly affects LC sovereign bond demand but does not fit any of the
↪  above. If you categorise it as such, explain what the mechanism could be.

Rules:
1. Emit one event per focal action. Read the cell holistically before splitting.
↪  "Date: M/D/YYYY" markers are dates, not necessarily event boundaries --
↪  multiple markers can describe aspects of the same action (one for announce,
↪  another for effective) or parts of a single policy package. Identify distinct
↪  focal actions first, then attach announce and effective dates from wherever
↪  they appear in the cell. When a cell opens with an announcement covering
↪  multiple related actions (a policy package -- same regulatory family,
↪  coordinated effective date), the opening announce date applies to all pieces
↪  of the package, not just the one explicitly named. Example: "Date: 6/10. It
↪  was announced RR will be raised, effective 6/24. Date: 6/24. Banks must also
↪  maintain additional securities." -- both pieces share announce=2022-06-10. A
↪  date or regulation mentioned in passing -- not as the cell's focus -- is
↪  context, not a separate event.
2. Emit every focal action, even if not repression-relevant. Set relevant=false
↪  for those, and leave direction and channel null. Do not silently drop
↪  non-relevant focal actions.
3. Use A sparingly - only when direction cannot be inferred (either the text does
↪  not reveal it, or the action is a non-change). Do not use A as a hedge when
↪  direction can be inferred from the wording.
4. Return announce_date and effective_date as YYYY-MM-DD ONLY when the text
↪  states a specific day. If the text gives only a month, quarter, or year,
↪  leave that field null. Do not infer day 01. Each date is evaluated
↪  independently. If the cell mentions the same action's date at multiple
↪  precisions (e.g., "announced in March 2013" and later "As announced March 3,
↪  2013"), use the most day-precise mention.
5. Reasoning must always be populated. When relevant=true, quote the phrase from
↪  the text that justifies BOTH the direction and the channel. When
↪  relevant=false, explain briefly why the action does not affect LC sovereign
↪  bond demand.
```

The cell-level user message follows the following template:

```
Country: {country}  ({iso3})
Instrument: {instrument}
Source month of the cell: {source_month_date}
```

```
Cell text:
---
{raw_cell_text}
---
```

Daily event panel. I construct the daily event panel used in Section 5.4 by retaining events with a day-precise announcement or effective date, mapping each event to the next available trading day in the DBP sample, and dropping any (currency, day) bin in which both U and D events appear. Relevant events with direction U or D supply the tightening and loosening indicators, and neutral events supply the $N_{j,t}$ indicator. Restricted to the 11 sample currencies and the period 2007–2024, the resulting panel contains 240 tightening and 97 loosening event-days. Table 8 summarises the daily event panel by country and direction, and reports the channel mix among the underlying U and D events.

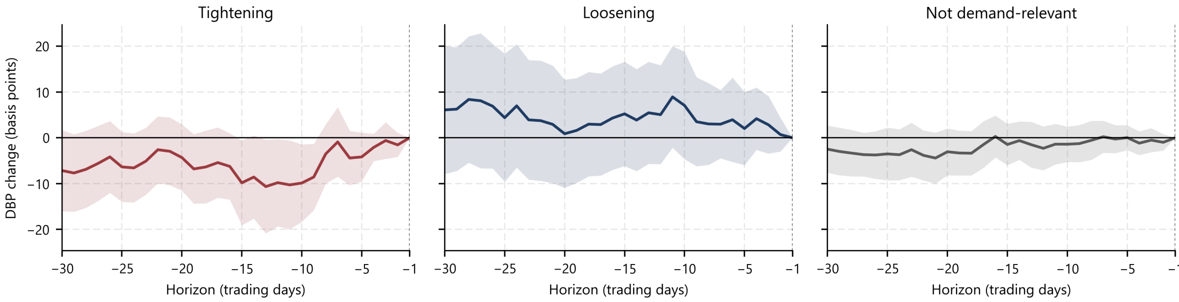
Table 8: Macroprudential policy events: country and direction

Country	Tightening (U)	Loosening (D)	Ambiguous (A)	Total
Brazil	11	18	3	32
China	45	19	0	64
Colombia	7	2	0	9
Hungary	16	2	3	21
Indonesia	19	7	2	28
India	22	11	0	33
Mexico	10	4	0	14
Poland	5	1	0	6
Russia	46	12	3	61
Turkey	52	20	0	72
South Africa	7	1	0	8
Total	**240**	**97**	**11**	**348**

Notes: Daily event-day counts from the collapsed event panel; restricted to the 11 sample currencies and 2007–2024. Underlying event-level channel mix among U and D events: mandate ($U = 134$, $D = 45$); indirect ($U = 264$, $D = 112$); foreign ($U = 7$, $D = 8$).

Pre-event dynamics. Figure 7 reports the pre-event panel local-projection estimates of $DBP_{j,\tau,t+h} - DBP_{j,\tau,t-1}$ for $h \in [-30, -1]$, with $h = -1$ normalised to zero. The specification uses only the contemporaneous treatment indicators and the currency–tenor–year and date fixed-effects structure of equation (32), without lagged ΔDBP, lagged treatment indicators, or lagged controls. For both event types, the pre-event coefficients are statistically indistinguishable from zero at most horizons. For tightening events, the mildly negative and statistically significant coefficients indicate that the DBP on the day before the event is, on average, slightly higher than over the preceding two weeks and not already drifting downward toward the post-event levels. The pre-event diagnostic therefore supports the interpretation that the post-event trends in Figure 3 reflect movement starting at the event date rather than a continuation of pre-existing DBP dynamics or anticipation effects.

Figure 7: Pre-event dynamics around macroprudential policy events



Notes: This figure plots pre-event panel local-projection estimates of changes in the DBP, in basis points, from 30 trading days before a macroprudential policy event through the day before the event. Tightening events (U, red, left panel) are actions predicted to increase government-bond demand; loosening events (D, blue, centre panel) are actions predicted to decrease demand; demand-neutral events (N, black, right panel) are classified as unrelated to government-bond demand. The dependent variable is $DBP_{j,\tau,t+h} - DBP_{j,\tau,t-1}$ for $h = -30, \ldots, -1$, with $h = -1$ normalised to zero. The specification includes contemporaneous event indicators and currency–tenor–year and date fixed effects, but excludes lagged ΔDBP, treatment indicators, and controls. Standard errors are Driscoll and Kraay (1998) with Bartlett kernel and horizon-dependent bandwidth $\max\{m(T), |h| + 1\}$, where $m(T) = \lfloor 4(T/100)^{2/9} \rfloor$. Shaded regions are 90% confidence bands. Sources: Bloomberg, Refinitiv, IMF, and author's calculations.